

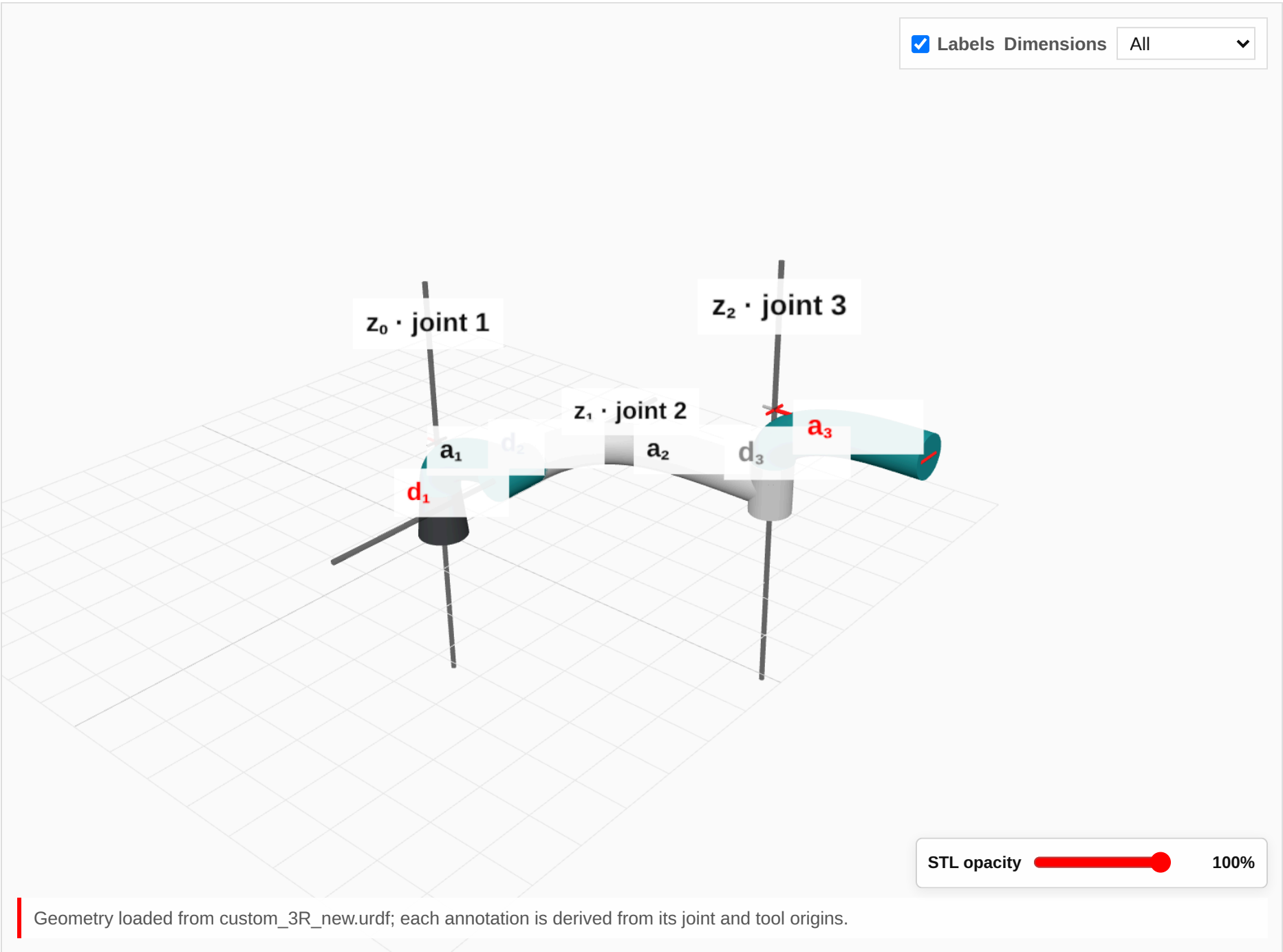
Inverse Kinematics of 3R Robots

Two 3R case studies reveal how robot geometry determines the inverse-kinematics method

First, derive a quartic for the offset-axis custom 3R robot; then exploit the intersecting axes with Paden–Kahan circle subproblems.

Kinematic geometry of the custom 3R robot

The lines connect consecutive D–H axes. Their signed lengths are the geometry used by both forward and inverse kinematics.



Inferred standard D–H geometry	
d_1	$0.5 + 0.5 = 1.00 \text{ m}$
a_1	1.00 m
d_2	1.25 m
a_2	2.00 m
d_3	0.75 m
a_3	1.50 m

The two consecutive base translations combine into $d_1 = 1.0 \text{ m}$; this translation commutes with $R_z(\theta_1)$.

Algebraic and geometric structure in 3R inverse kinematics

Both methods enumerate branches analytically; they differ in what they exploit.

01

Algebraic · Pieper

Use standard D–H parameters, eliminate θ_1 , eliminate θ_2 , and solve one quartic in $t_3 = \tan(\theta_3/2)$.

 $\leftrightarrow$ **02**

Geometric · Paden–Kahan

View a revolute joint as a circle, then solve circle–point and circle–circle constraints.

Learning goal: recognize when elimination is generic—and when robot geometry gives a shorter route.

Sources: D. L. Pieper, *The Kinematics of Manipulators under Computer Control*, 1968; W. Kahan, *Lectures on Computational Aspects of Geometry*, 1983; B. E. Paden, *Kinematics and Control of Robot Manipulators*, 1986; R. Murray, Z. Li, and S. Sastry, *A Mathematical Introduction to Robotic Manipulation*, 1994.

PROBLEM SETUP

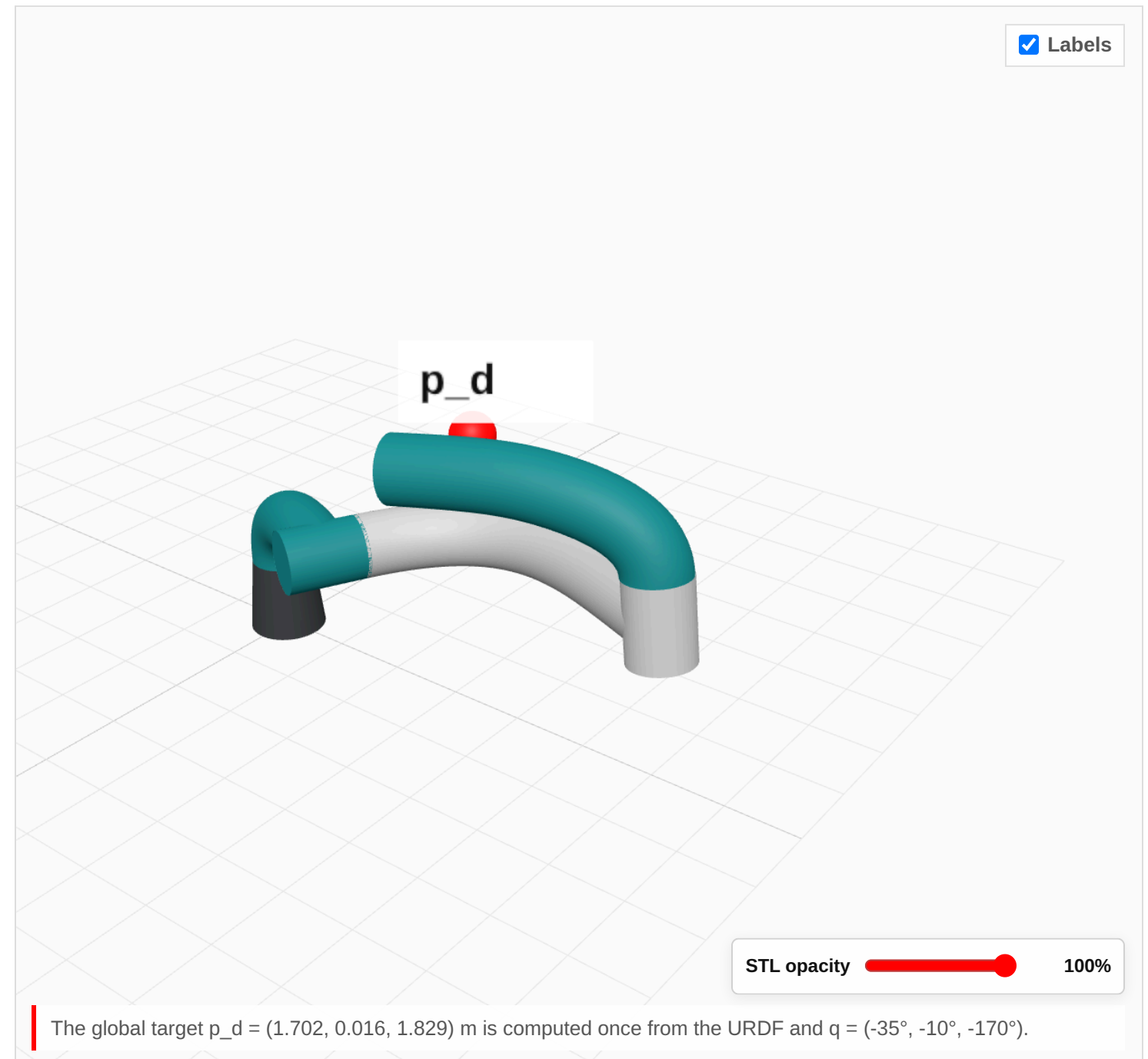
The 3R position inverse-kinematics problem

Given an end-effector position $\mathbf{p}_d = [x \ y \ z_d]^T$, find every $\boldsymbol{\theta} = [\theta_1 \ \theta_2 \ \theta_3]^T$ that produces it—or establish that none exists.

$$\mathbf{p}(\theta_1, \theta_2, \theta_3) = \mathbf{p}_d$$

$$\mathbf{q} = (-35^\circ, -10^\circ, -170^\circ) \Rightarrow \mathbf{p}_d = \begin{bmatrix} 1.7018 \\ 0.0164 \\ 1.8294 \end{bmatrix} \text{ m}$$

Input	D–H geometry and $\mathbf{p}_d$
Output	all real joint triples within limits
Branches	one position may have several IK solutions



One target can have several—or no—IK solutions

Forward kinematics maps each joint vector to one position; inverse kinematics asks for the entire set of joint vectors that reach a target.

Empty

No joint vector reaches the target.

Finite and multiple

Several isolated joint vectors reach the same target.

Degenerate

Branches may merge at an isolated solution; special geometry may instead allow a continuum.

Function

$f : D \rightarrow Y$ assigns **one output** to every input in D .

Image

$f(D) = \{f(t) : t \in D\}$: all outputs the function actually reaches.

Preimage of a target

$f^{-1}(\{y\}) = \{t \in D : f(t) = y\}$: **all inputs** that reach y .

Example: $f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(t) = \max(t^2 - 1, 0)$

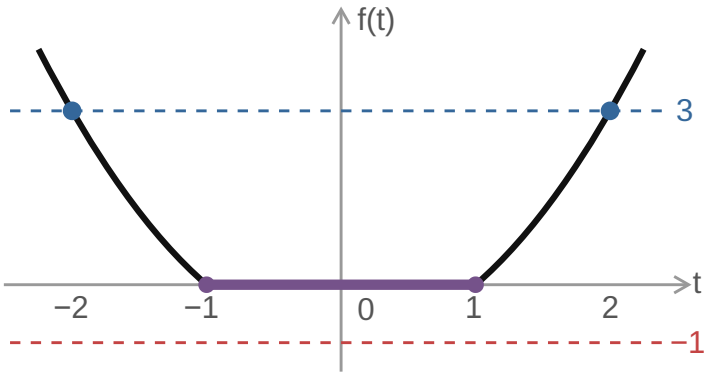


Image: $f(\mathbb{R}) = [0, \infty)$.

Target y	Preimage $f^{-1}(\{y\})$	What happens?
-1	$\emptyset$	Empty: below the image.
3	$\{-2, 2\}$	Finite and multiple: two inputs.
0	$[-1, 1]$	Degenerate: a whole interval.

A horizontal line finds the preimage. The flat segment illustrates a continuum; merging isolated branches need not create one.

Standard D–H convention for the 3R chain

We use standard D–H transforms. This fixes the meaning of every parameter in the derivation.

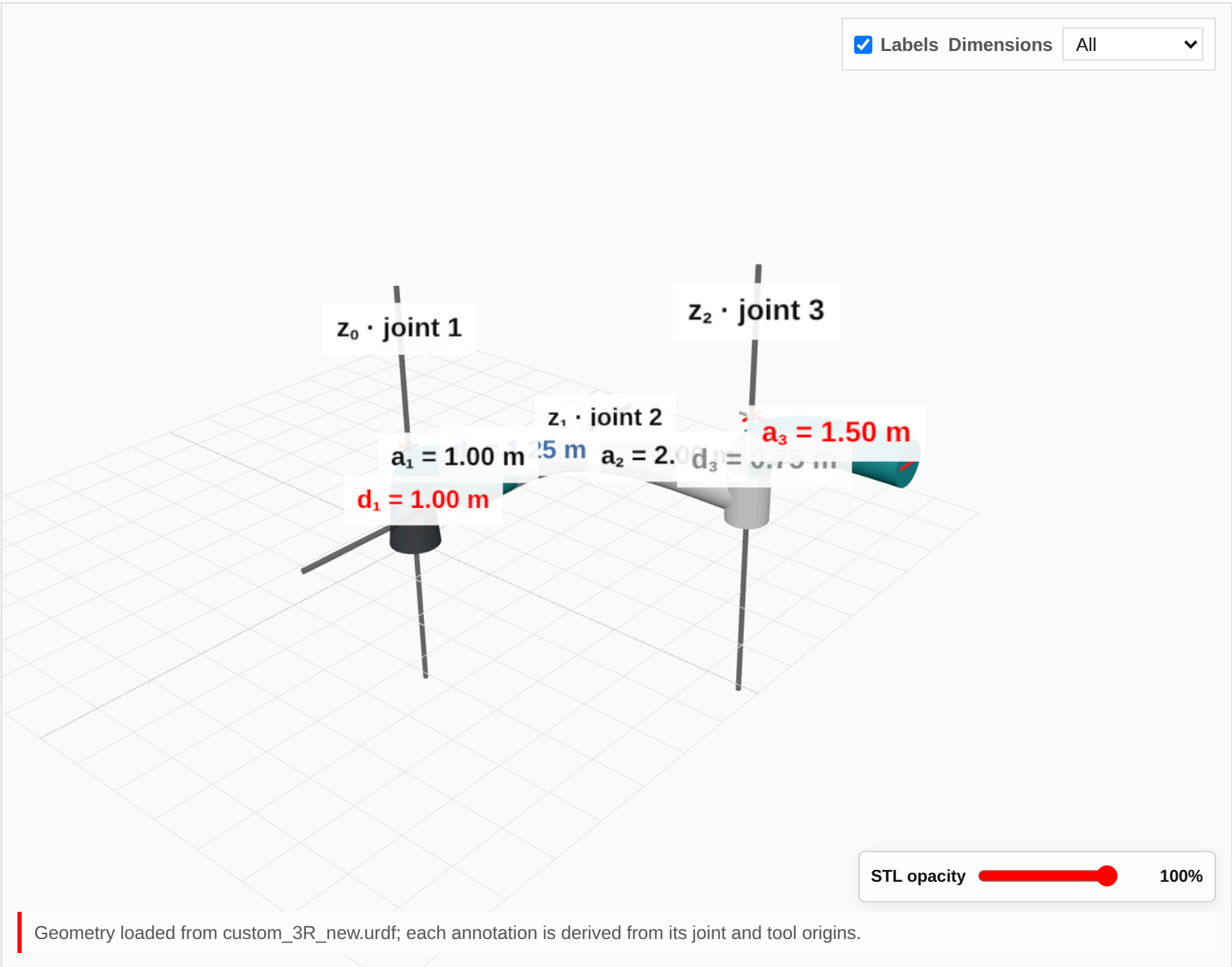
$${}^{i-1}T_i = R_z(\theta_i)T_z(d_i)T_x(a_i)R_x(\alpha_i)$$

joint i	a_i	α_i	d_i	variable
1	a_1	α_1	d_1	θ_1
2	a_2	α_2	d_2	θ_2
3	a_3	α_3	d_3	θ_3

Position only: α_3 changes the terminal frame orientation but not the origin O_3 , so it does not enter $\mathbf{p}_{O_3}$.

D-H parameters of the custom 3R robot

Placing F_0 at the world origin and absorbing the vertical offset into d_1 reproduces the model geometry exactly.



i	a_i	α_i	d_i
1	1.00	-90°	1.00
2	2.00	$+90^\circ$	1.25
3	1.50	0°	0.75

$$\mathbf{p}(-35^\circ, -10^\circ, -170^\circ) = \begin{bmatrix} 1.7018 \\ 0.0164 \\ 1.8294 \end{bmatrix} \text{ m}$$

This position becomes the numerical IK target for the remainder of the algebraic method.

Two invariants remove the first two joint angles

For the custom robot, $\alpha_1 = -\pi/2$, $\alpha_2 = \pi/2$. Write $c_i = \cos \theta_i$, $s_i = \sin \theta_i$.

First remove the base offset: $z := z_d - d_1$. Then $R := x^2 + y^2 + z^2$ and z are unchanged by rotation about joint 1, so θ_1 disappears.

Write the invariants as linear polynomials in c_2, s_2

$$\begin{aligned} R &= 2a_1(F_1c_2 + F_2s_2) + F_3, \\ z &= F_2c_2 - F_1s_2. \end{aligned}$$

$$\begin{aligned} F_1 &= a_2 + a_3c_3, & F_2 &= d_3, & C &= d_2 + a_3s_3, \\ F_3 &= a_1^2 + F_1^2 + F_2^2 + C^2 \\ &= a_1^2 + a_2^2 + a_3^2 + d_2^2 + d_3^2 \\ &\quad + 2a_2a_3c_3 + 2d_2a_3s_3. \end{aligned}$$

Each F_i is constant or affine in c_3, s_3 .

Eliminate c_2, s_2 , then solve for θ_3

$$E := \frac{R - F_3}{2a_1} = F_1c_2 + F_2s_2$$

Square and add the equations for E and z : the cross terms cancel and $c_2^2 + s_2^2 = 1$.

$$E^2 + z^2 = F_1^2 + F_2^2.$$

$$t_3 = \tan \frac{\theta_3}{2}, \quad c_3 = \frac{1 - t_3^2}{1 + t_3^2}, \quad s_3 = \frac{2t_3}{1 + t_3^2}$$

Substitute and clear $(1 + t_3^2)^2$: a polynomial of degree at most four determines θ_3 .

Here z_d is world height, z is shifted height, and R is a squared distance. Solve the quartic, then recover the other two angles by linear algebra.

Once θ_3 is known, recover θ_2 and θ_1

Evaluate F_1, F_2, F_3, C at each real third-joint candidate. Cramer's rule solves two 2×2 systems.

1 · Obtain c_2, s_2 from R, z

$$\begin{bmatrix} 2a_1F_1 & 2a_1F_2 \\ F_2 & -F_1 \end{bmatrix} \begin{bmatrix} c_2 \\ s_2 \end{bmatrix} = \begin{bmatrix} R - F_3 \\ z \end{bmatrix}$$

$$E = \frac{R - F_3}{2a_1}, \quad \Delta_2 = F_1^2 + F_2^2$$

$$c_2 = \frac{F_1 E + F_2 z}{\Delta_2},$$

$$s_2 = \frac{F_2 E - F_1 z}{\Delta_2},$$

$$\theta_2 = \text{atan2}(s_2, c_2).$$

2 · Obtain c_1, s_1 from x, y

$$A = a_1 + F_1 c_2 + F_2 s_2, \quad C = d_2 + a_3 s_3$$

$$\begin{bmatrix} A & -C \\ C & A \end{bmatrix} \begin{bmatrix} c_1 \\ s_1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \Delta_1 = A^2 + C^2$$

$$c_1 = \frac{Ax + Cy}{\Delta_1},$$

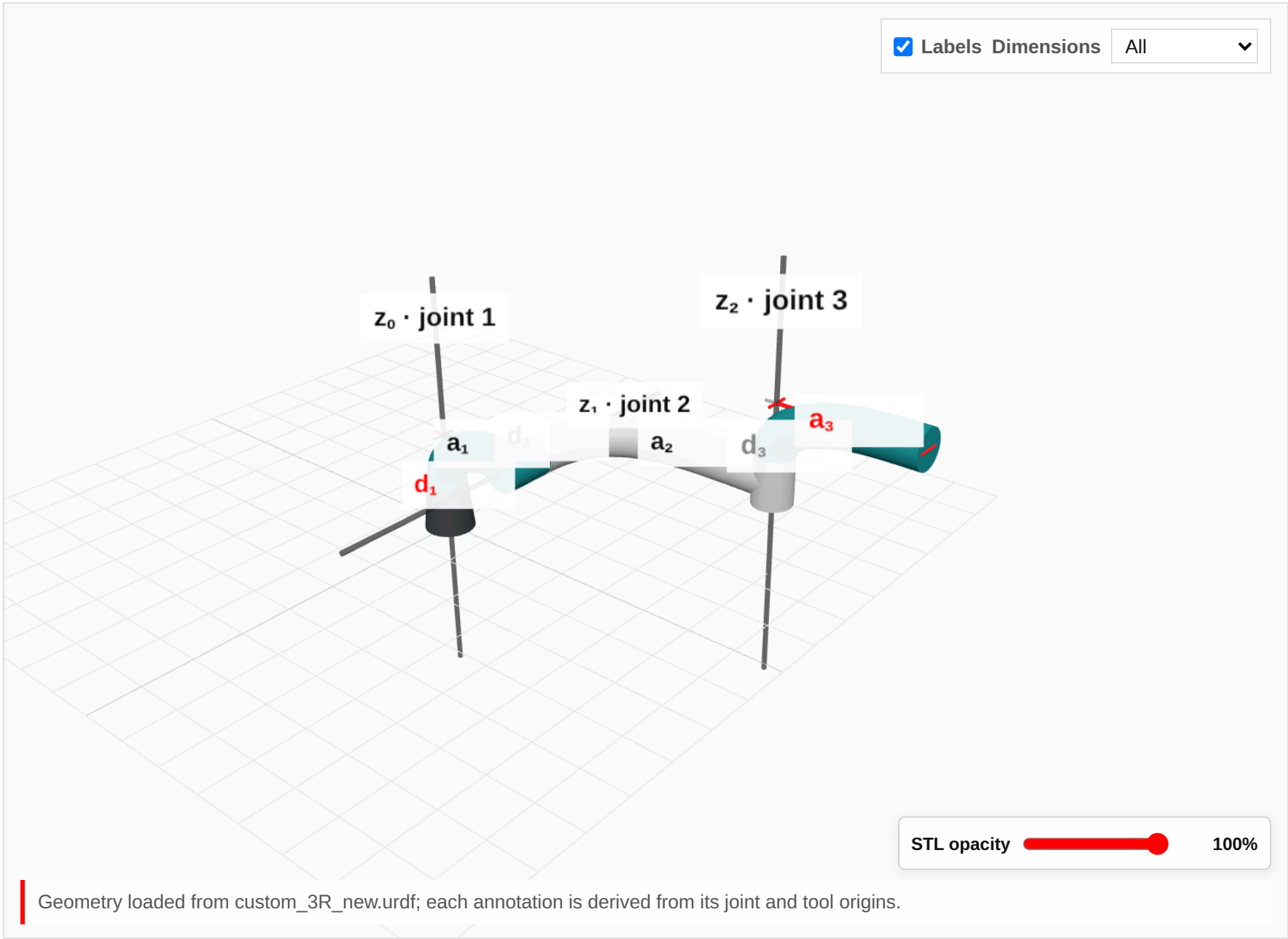
$$s_1 = \frac{Ay - Cx}{\Delta_1},$$

$$\theta_1 = \text{atan2}(s_1, c_1).$$

Use these formulas when $a_1 \neq 0$, $\Delta_2 \neq 0$, and $\Delta_1 \neq 0$. For every branch, enforce joint limits and verify the original world target $[x, y, z_d]^T$ by forward kinematics.

Start with the custom 3R robot at zero joint angles

Use `custom_3R_new.urdf`, the offset-axis robot introduced earlier. The base frame $\{0\}$ coincides with the world frame.



Read the home geometry

At $\theta = \mathbf{0}$, the joint axes point along $+z, +y, +z$.

$${}^0\omega_1 = \mathbf{e}_z, \quad {}^0\omega_2 = \mathbf{e}_y, \quad {}^0\omega_3 = \mathbf{e}_z.$$

The D–H table supplies the axis offsets and the tool’s home position.

$$\mathbf{p}_0 = \begin{bmatrix} a_1 + a_2 + a_3 \\ d_2 \\ d_1 + d_3 \end{bmatrix}$$

All three joint variables are zero here. The tool’s home orientation is I ; its home pose is $M = [I, \mathbf{p}_0; \mathbf{0}^T, 1]$.

Express forward kinematics as a product of exponentials

Each constant screw ${}^0S_i = ({}^0\boldsymbol{\omega}_i, {}^0\mathbf{v}_i)$ describes joint i at home, expressed in the same fixed base frame.

$${}^0T_{ee}(\boldsymbol{\theta}) = e^{[{}^0S_1]\theta_1} e^{[{}^0S_2]\theta_2} e^{[{}^0S_3]\theta_3} M, \quad [{}^0S_i] = \begin{bmatrix} [{}^0\boldsymbol{\omega}_i]_{\times} & {}^0\mathbf{v}_i \\ \mathbf{0}^T & 0 \end{bmatrix}.$$

Choose a point ${}^0\mathbf{r}_i$ on each home axis. For a revolute joint, ${}^0\mathbf{v}_i = -{}^0\boldsymbol{\omega}_i \times {}^0\mathbf{r}_i$.

Joint	${}^0\boldsymbol{\omega}_i$	Point ${}^0\mathbf{r}_i$ on the axis	${}^0\mathbf{v}_i$
1	$(0, 0, 1)$	$(0, 0, 0)$	$(0, 0, 0)$
2	$(0, 1, 0)$	$(a_1, 0, d_1)$	$(-d_1, 0, a_1)$
3	$(0, 0, 1)$	$(a_1 + a_2, d_2, d_1)$	$(d_2, -a_1 - a_2, 0)$

The screws encode axis direction *and* location. The rightmost factor M supplies the tool’s home pose; the full product maps tool coordinates into base coordinates.

The D–H geometry gives three numerical screws and M

$(a_1, a_2, a_3) = (1, 2, 1.5) \text{ m}$, $(d_1, d_2, d_3) = (1, 1.25, 0.75) \text{ m}$, and $(\alpha_1, \alpha_2, \alpha_3) = (-90^\circ, 90^\circ, 0^\circ)$.

Stack angular then linear components

$${}^0S_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad {}^0S_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \quad {}^0S_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1.25 \\ -3 \\ 0 \end{bmatrix}.$$

For example, $-\mathbf{e}_y \times (1, 0, 1) = (-1, 0, 1)$ gives the linear part of 0S_2 .

Substitute the home tool pose

$$M = \begin{bmatrix} 1 & 0 & 0 & 4.5 \\ 0 & 1 & 0 & 1.25 \\ 0 & 0 & 1 & 1.75 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

$$\begin{bmatrix} \mathbf{p}(\boldsymbol{\theta}) \\ 1 \end{bmatrix} = e^{[{}^0S_1]\theta_1} e^{[{}^0S_2]\theta_2} e^{[{}^0S_3]\theta_3} \begin{bmatrix} 4.5 \\ 1.25 \\ 1.75 \\ 1 \end{bmatrix}.$$

PoE and the standard D–H product describe the same robot. The values above come from the home axes, not from the axes after the robot has moved.

Expand PoE into $c_1, s_1, c_2, s_2, c_3, s_3$

Write $c_i = \cos \theta_i, s_i = \sin \theta_i$. A revolute exponential acts on a point as ${}^0\mathbf{r}_i + R_{\omega_i}(\theta_i)(\mathbf{p} - {}^0\mathbf{r}_i)$.

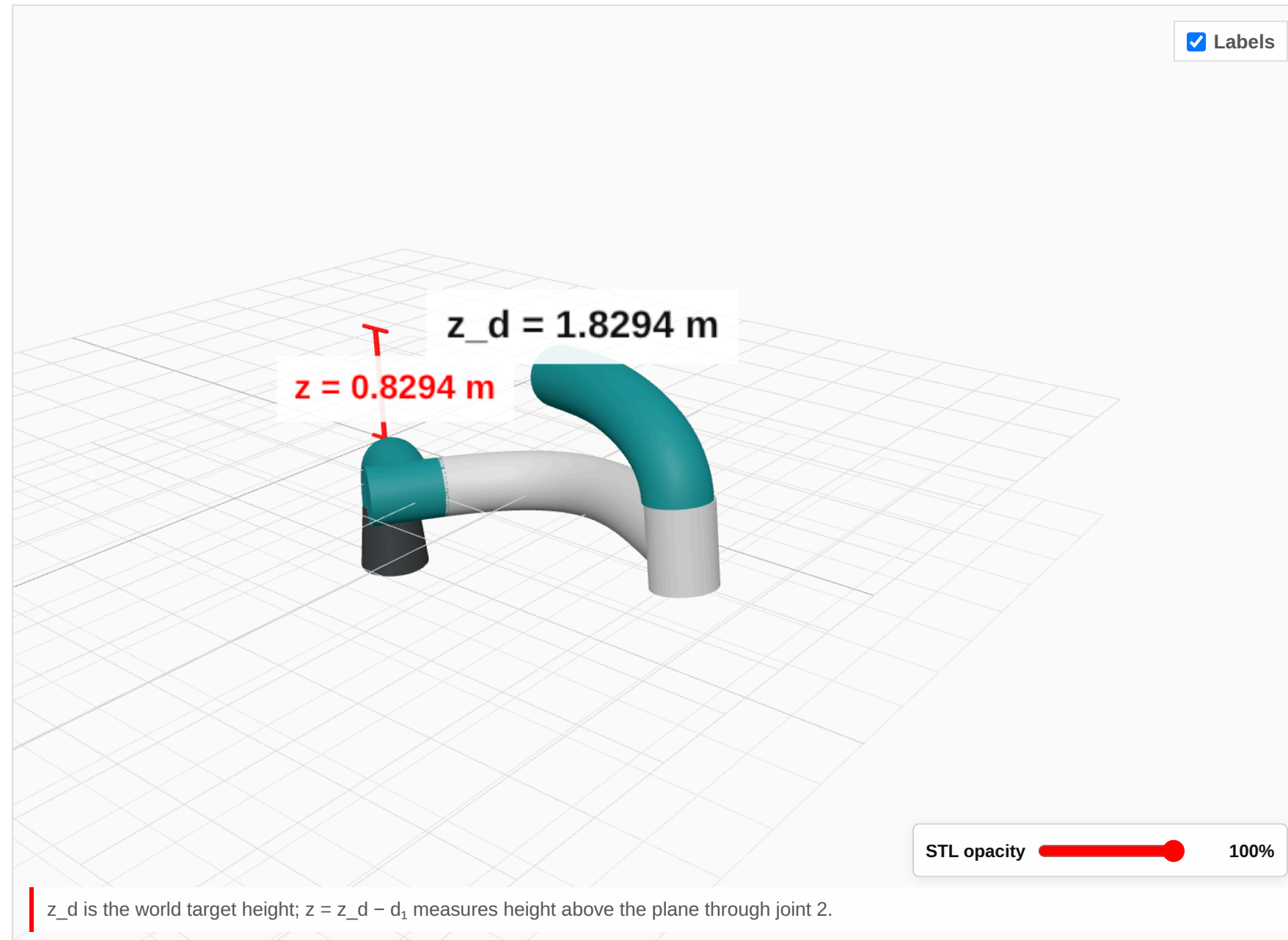
$$\mathbf{p}(\boldsymbol{\theta}) = R_z(\theta_1) \begin{bmatrix} 1 + (2 + 1.5c_3)c_2 + 0.75s_2 \\ 1.25 + 1.5s_3 \\ 1 + 0.75c_2 - (2 + 1.5c_3)s_2 \end{bmatrix}.$$

$$\begin{aligned} x &= c_1 + 2c_1c_2 + 1.5c_1c_2c_3 + 0.75c_1s_2 - 1.25s_1 - 1.5s_1s_3, \\ y &= s_1 + 2s_1c_2 + 1.5s_1c_2c_3 + 0.75s_1s_2 + 1.25c_1 + 1.5c_1s_3, \\ z_d &= 1 + 0.75c_2 - 2s_2 - 1.5s_2c_3. \end{aligned}$$

These are the three equations to solve for a given world target $\mathbf{p}_d = [x, y, z_d]^T$. Keep z_d for world height; next we subtract the fixed base offset.

The height equation contains no c_1 or s_1

Joint 1 rotates about the world vertical axis. It changes azimuth, while leaving the vertical coordinate unchanged.



Subtract the base offset once

$$z_d = 1 + 0.75c_2 - (2 + 1.5c_3)s_2.$$

$$\begin{aligned} z &:= z_d - d_1 = z_d - 1, \\ &= 0.75c_2 - (2 + 1.5c_3)s_2. \end{aligned}$$

Only θ_2, θ_3 remain.

Both z_d and z are independent of θ_1 . The shifted height z will simplify the squared-distance equation.

In $x^2 + y^2$, the first-joint terms cancel

Collect the parts of the coordinate equations that depend only on θ_2, θ_3 .

$$A := 1 + (2 + 1.5c_3)c_2 + 0.75s_2, \quad C := 1.25 + 1.5s_3.$$

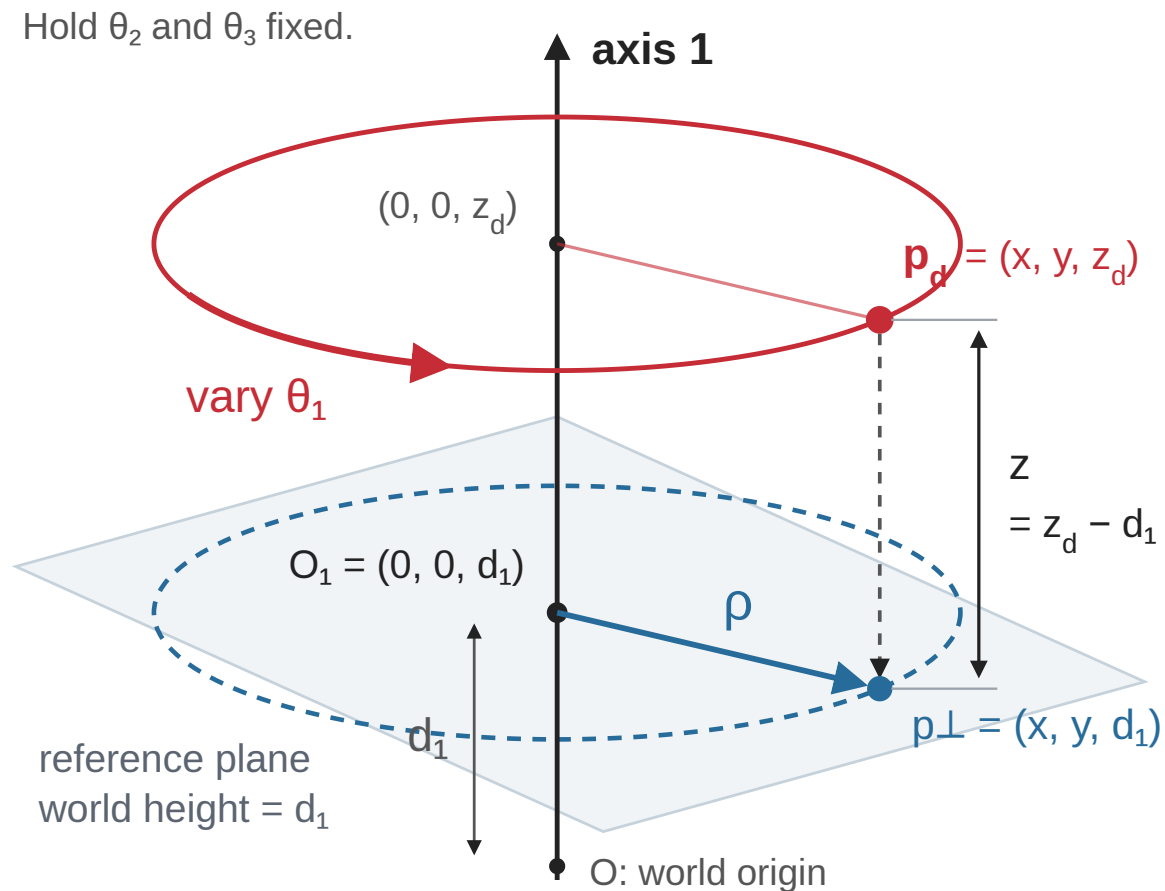
$$x = Ac_1 - Cs_1, \quad y = As_1 + Cc_1.$$

$$\begin{aligned} x^2 + y^2 &= (Ac_1 - Cs_1)^2 + (As_1 + Cc_1)^2 \\ &= A^2(c_1^2 + s_1^2) + C^2(s_1^2 + c_1^2) \underbrace{-2ACc_1s_1 + 2ACs_1c_1}_0 \\ &= A^2 + C^2. \end{aligned}$$

Define $\rho^2 := x^2 + y^2 = A^2 + C^2$. Rotation about joint 1 preserves distance to its axis, so ρ is independent of θ_1 .

Joint 1 moves the end effector along a circle around its axis

Hold θ_2, θ_3 fixed and vary only θ_1 . The circle through $\mathbf{p}_d$ has constant height and constant radius.



Project onto the plane at height d_1

$$O_1 = (0, 0, d_1), \quad \mathbf{p}_\perp = (x, y, d_1).$$

$$\|\mathbf{p}_\perp - O_1\| = \rho = \sqrt{x^2 + y^2},$$

$$z_d - d_1 = z.$$

The upper circle is centered at $(0, 0, z_d)$ on axis 1. Its projection is a circle of the same radius centered at O_1 .

Changing θ_1 moves the point around these circles. It changes neither ρ nor z .

Use $R = x^2 + y^2 + z^2$ and z to remove θ_1

$R = \|\mathbf{p}_d - (0, 0, d_1)^T\|^2$ is a squared distance from a fixed point on axis 1. Here $z = z_d - d_1$, not world height.

$$R = \rho^2 + z^2 = A^2 + C^2 + z^2.$$

The terms quadratic in c_2, s_2 also simplify:

$$\underbrace{\left[(2 + 1.5c_3)c_2 + 0.75s_2\right]^2}_{(A-1)^2} + \underbrace{\left[0.75c_2 - (2 + 1.5c_3)s_2\right]^2}_{z^2} = (2 + 1.5c_3)^2 + 0.75^2.$$

$$\begin{aligned} R &= 2\left[(2 + 1.5c_3)c_2 + 0.75s_2\right] \\ &\quad + 1 + (2 + 1.5c_3)^2 + 0.75^2 + (1.25 + 1.5s_3)^2, \\ z &= 0.75c_2 - (2 + 1.5c_3)s_2. \end{aligned}$$

Both equations are independent of θ_1 , and now both are linear in c_2, s_2 .

Collect the third-joint terms into F_1, F_2, F_3

The coefficient functions are obtained directly from the preceding equations. They contain only θ_3 and fixed geometry.

$$F_1 := a_2 + a_3 c_3 = 2 + 1.5 c_3,$$

$$F_2 := d_3 = 0.75,$$

$$C = d_2 + a_3 s_3 = 1.25 + 1.5 s_3,$$

$$F_3 := a_1^2 + F_1^2 + F_2^2 + C^2.$$

$$\begin{aligned} F_3 &= 1 + 4 + 0.5625 + 1.5625 \\ &\quad + 2.25(c_3^2 + s_3^2) + 6c_3 + 3.75s_3 \\ &= 9.375 + 6c_3 + 3.75s_3. \end{aligned}$$

The two invariant equations

$$\begin{aligned} R &= 2a_1(F_1 c_2 + F_2 s_2) + F_3, \\ z &= F_2 c_2 - F_1 s_2. \end{aligned}$$

Here $a_1 = 1$. Every F_i is constant or affine in c_3, s_3 .

For a given target, R, z are known. The remaining unknowns are θ_2 and θ_3 .

Eliminate s_2 , then use $c_2^2 + s_2^2 = 1$

Set $E = (R - F_3)/(2a_1)$. The equations are $E = F_1c_2 + F_2s_2$ and $z = F_2c_2 - F_1s_2$.

$$\begin{aligned} F_1E + F_2z &= (F_1^2 + F_2^2)c_2 && \text{(the } s_2 \text{ terms cancel),} \\ F_2E - F_1z &= (F_1^2 + F_2^2)s_2 && \text{(the } c_2 \text{ terms cancel).} \end{aligned}$$

Squaring and adding the original two equations gives the same closure directly:

$$E^2 + z^2 = (F_1^2 + F_2^2)(c_2^2 + s_2^2) = F_1^2 + F_2^2.$$

$$(R - F_3)^2 + 4a_1^2z^2 - 4a_1^2(F_1^2 + F_2^2) = 0.$$

Both θ_1 and θ_2 are gone. Substituting F_1, F_2, F_3 leaves an equation quadratic in c_3, s_3 .

Substitute $t_3 = \tan(\theta_3/2)$ to obtain a quartic

Let $H := R - 9.375$. The custom robot's closure is
 $(H - 6c_3 - 3.75s_3)^2 + 4z^2 - 4(2 + 1.5c_3)^2 - 2.25 = 0$.

$$c_3 = \frac{1 - t_3^2}{1 + t_3^2}, \quad s_3 = \frac{2t_3}{1 + t_3^2}, \quad \theta_3 = 2 \operatorname{atan}(t_3).$$

Substitute and multiply by $(1 + t_3^2)^2$:

$$P(t_3) = [(H + 6)t_3^2 - 7.5t_3 + (H - 6)]^2 + (4z^2 - 2.25)(1 + t_3^2)^2 - 4(3.5 + 0.5t_3^2)^2 = 0.$$

$$\begin{aligned} b_4 &= (H + 6)^2 + 4z^2 - 3.25, \\ b_3 &= -15(H + 6), \\ b_2 &= 2H^2 + 8z^2 - 34.25, \end{aligned}$$

$$\begin{aligned} b_1 &= -15(H - 6), \\ b_0 &= (H - 6)^2 + 4z^2 - 51.25, \\ P(t_3) &= b_4t_3^4 + b_3t_3^3 + b_2t_3^2 + b_1t_3 + b_0. \end{aligned}$$

What do the roots of the quartic tell us?

If $b_4 \neq 0$, $P(t_3)$ has four complex roots counting multiplicity. Its real coefficients make nonreal roots occur in conjugate pairs.

Real roots give angle candidates

A generic quartic has 0, 2, or 4 real roots. Each real $t_{3,j}$ gives $\theta_{3,j} = 2 \operatorname{atan}(t_{3,j})$, modulo 2π .

Repeated roots can merge branches

A repeated root counts more than once algebraically. The number of *distinct* real candidates can therefore be 1 or 3 as well.

Check the missing half-angle point

$\theta_3 = \pi$ is not represented by finite t_3 . Test $(c_3, s_3) = (-1, 0)$ in the closure; here this condition is $b_4 = 0$. Then the finite polynomial has lower degree.

Solve, recover, then validate

Quartics admit formulas in radicals. In computation, use a polynomial root solver, recover θ_2, θ_1 , enforce joint limits, and check forward kinematics.

A real root is a third-joint candidate. The complete IK answer is the set of joint triples that survive back-substitution and validation.

Insert $\mathbf{p}_d$ and solve the numerical quartic

Use the target from the problem setup. It was generated at $(-35^\circ, -10^\circ, -170^\circ)$; IK must recover all configurations reaching it.

$$\mathbf{p}_d \simeq \begin{bmatrix} 1.701776 \\ 0.016394 \\ 1.829387 \end{bmatrix} \text{ m.}$$

$$\begin{aligned} z &= z_d - 1 \simeq 0.829387 \text{ m,} \\ \rho^2 &= x^2 + y^2 \simeq 2.896310 \text{ m}^2, \\ R &= \rho^2 + z^2 \simeq 3.584193 \text{ m}^2, \\ H &= R - 9.375 \simeq -5.790807. \end{aligned}$$

Evaluate $b_4, \dots, b_0$

$$\begin{aligned} P(t_3) &\simeq -0.454707t_3^4 - 3.137889t_3^3 \\ &\quad + 38.319963t_3^2 + 176.862111t_3 \\ &\quad + 90.524670 = 0. \end{aligned}$$

The four real roots are approximately

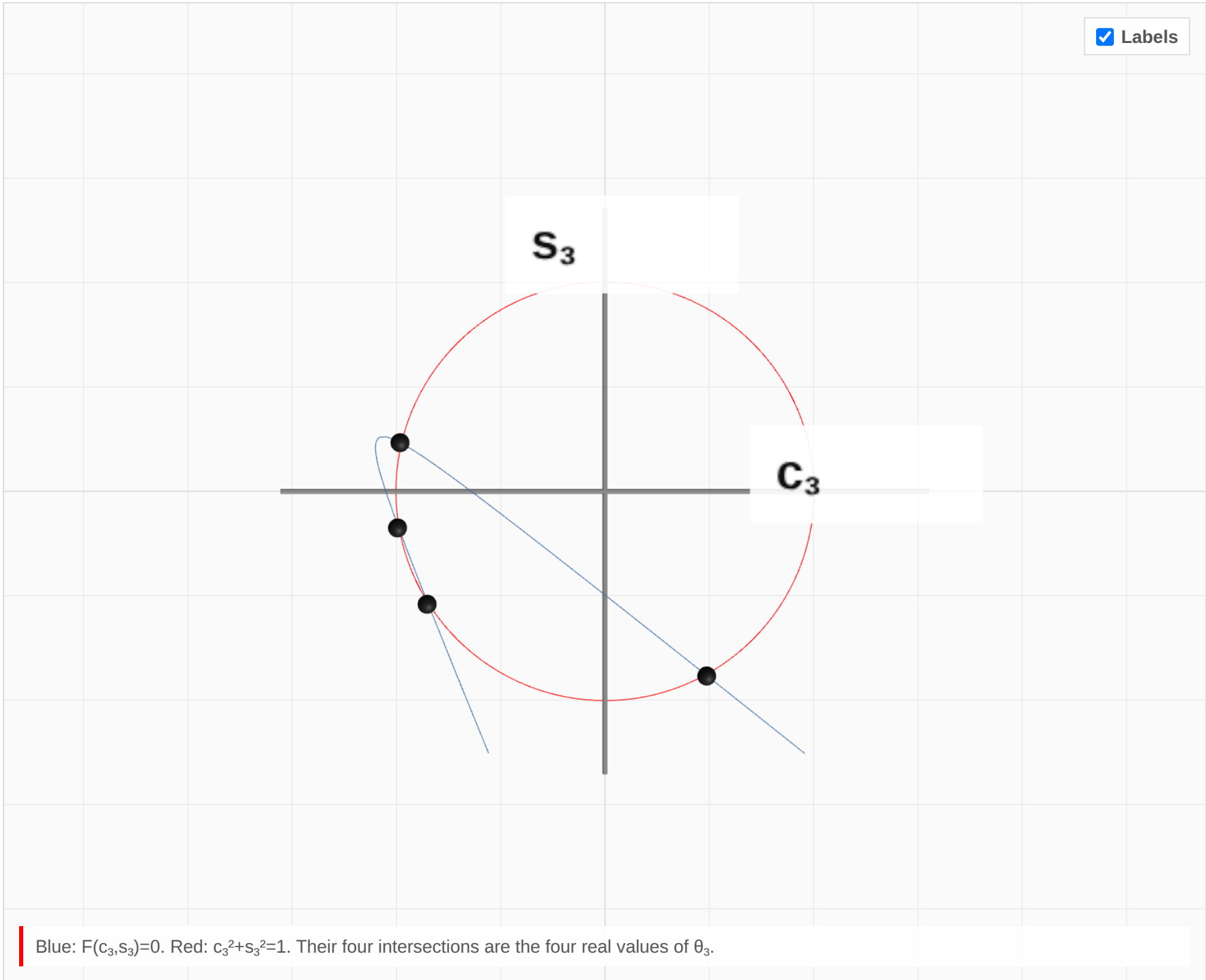
$$t_{3,j} \in \{-11.430052, -3.442869, -0.590812, 8.562827\}.$$

The coefficients and roots use the full-precision target; the displayed numbers are rounded.

Convert each root to θ_3 , then return to the two linear systems to obtain θ_2 and θ_1 .

Conic–circle intersections give the third-joint candidates

Substituting the custom robot’s F_i , target invariant R , and shifted height z gives the numerical conic.



Substitute the custom 3R parameters

$$(a_1, a_2, a_3, d_2, d_3) = (1, 2, 1.5, 1.25, 0.75),$$
$$R = 3.584193,$$
$$z = 0.829387.$$

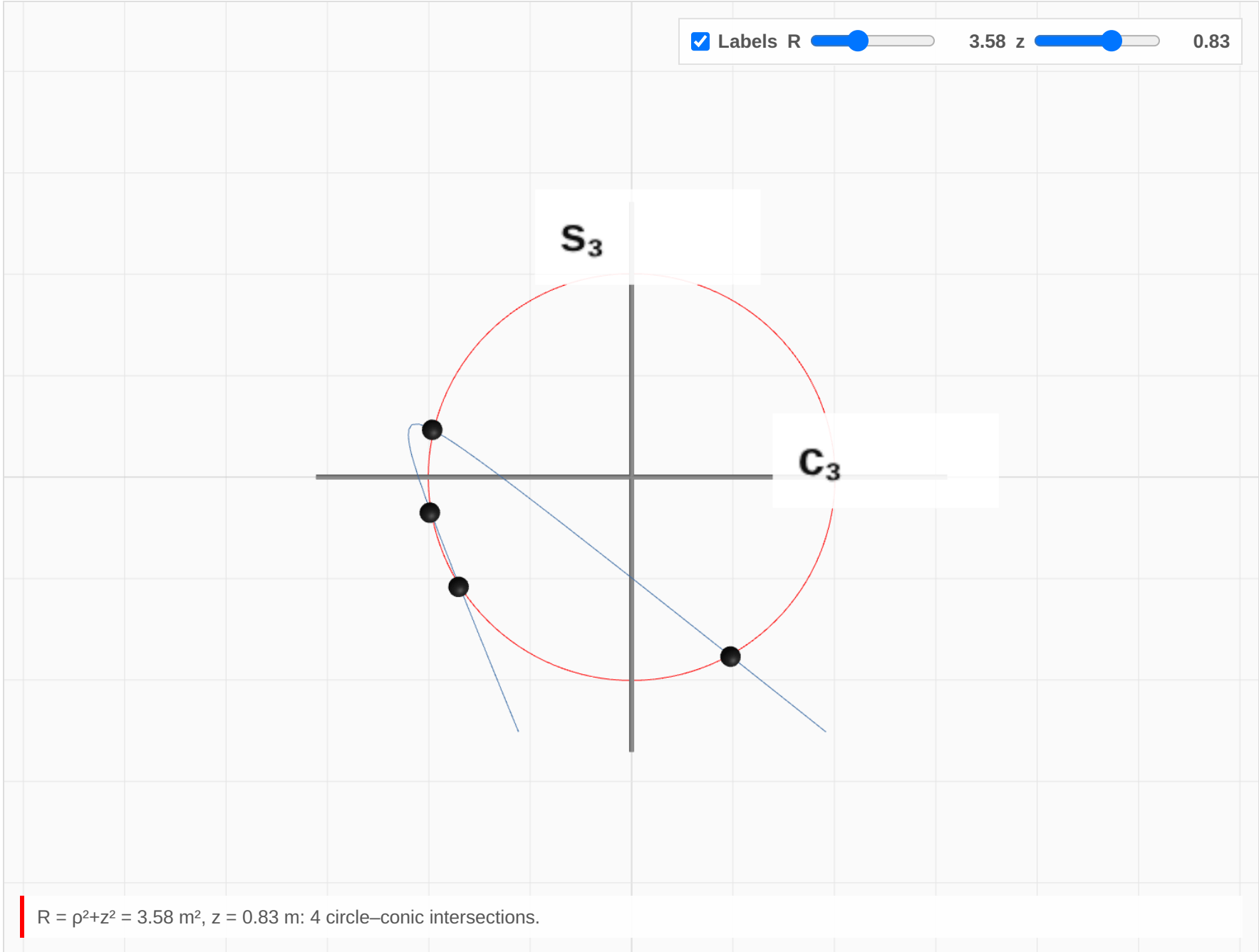
$$E = \frac{-5.790807 - 6c_3 - 3.75s_3}{2},$$
$$E^2 + 0.829387^2 - (2 + 1.5c_3)^2 - 0.75^2 = 0,$$

$$6.75c_3^2 + 11.25c_3s_3 + 3.515625s_3^2$$
$$+ 11.372422c_3 + 10.857764s_3$$
$$+ 4.508745 = 0$$

Together with $c_3^2 + s_3^2 - 1 = 0$, this conic yields four real values $\theta_3 = \text{atan2}(s_3, c_3)$.

Target invariants reshape the third-joint conic

Changing the target invariants $R = \rho^2 + z^2$ and z reshapes the conic while the trigonometric unit circle remains fixed.



The controls update

$$\begin{aligned} F_1 &= 2 + 1.5c_3, \\ F_2 &= 0.75, \\ F_3 &= 9.375 + 6c_3 + 3.75s_3, \\ E &= \frac{R - F_3}{2}, \\ F(c_3, s_3) &= E^2 + z^2 - F_1^2 - F_2^2 = 0. \end{aligned}$$

The red unit circle represents $c_3^2 + s_3^2 = 1$
Black intersection markers are recomputed as the target invariants change.

The number of intersections is the number of real third-joint candidates before back-substitution.

Four real branches of the numerical IK problem

The quartic has four real roots, and back-substitution maps each root to a distinct joint configuration at the same target position.

☒ Labels ☒ IKS 1 ☐ IKS 2 ☐ IKS 3 ☐ IKS 4

same p_d

IKS 1

STL opacity

100%

Each switch independently preserves an IK branch, so any subset of the four solutions can be compared at the common target.

Equivalently, scale the preceding polynomial by $-1/4$:

$$0.113677t_3^4 + 0.784472t_3^3 - 9.579991t_3^2 - 44.215528t_3 - 22.631168 = 0.$$

$t_{3,1}$

$-11.430052 \Rightarrow \theta_3 = -170.000^\circ$

$t_{3,2}$

$-3.442869 \Rightarrow \theta_3 = -147.608^\circ$

$t_{3,3}$

$-0.590812 \Rightarrow \theta_3 = -61.150^\circ$

$t_{3,4}$

$8.562827 \Rightarrow \theta_3 = 166.678^\circ$

Use the solution switches to retain any subset of the four configurations at the common red target point.

Recover the second joint from the same two equations

At a real θ_3 , evaluate the coefficients and solve for the two trigonometric unknowns.

$$\underbrace{\begin{bmatrix} F_1 & F_2 \\ F_2 & -F_1 \end{bmatrix}}_K \begin{bmatrix} c_2 \\ s_2 \end{bmatrix} = \begin{bmatrix} E \\ z \end{bmatrix}, \quad \det K = -(F_1^2 + F_2^2).$$

$$\begin{aligned} c_2 &= \frac{\det \begin{bmatrix} E & F_2 \\ z & -F_1 \end{bmatrix}}{\det K} = \frac{F_1 E + F_2 z}{F_1^2 + F_2^2}, \\ s_2 &= \frac{\det \begin{bmatrix} F_1 & E \\ F_2 & z \end{bmatrix}}{\det K} = \frac{F_2 E - F_1 z}{F_1^2 + F_2^2}, \\ \theta_2 &= \text{atan2}(s_2, c_2). \end{aligned}$$

Closure

$E^2 + z^2 = F_1^2 + F_2^2$ guarantees a unit (c_2, s_2) .

Invertibility

Here $F_2 = 0.75$, so $F_1^2 + F_2^2 > 0$ for every θ_3 .

Cramer’s solution also has a planar-angle interpretation

Using physical shifted height z , the vector (F_1, F_2) rotates into (E, z) through $-\theta_2$ in the displayed x – z plane.

$$\begin{bmatrix} E \\ z \end{bmatrix} = \begin{bmatrix} c_2 & s_2 \\ -s_2 & c_2 \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

The two vectors have the same length by the quartic closure.

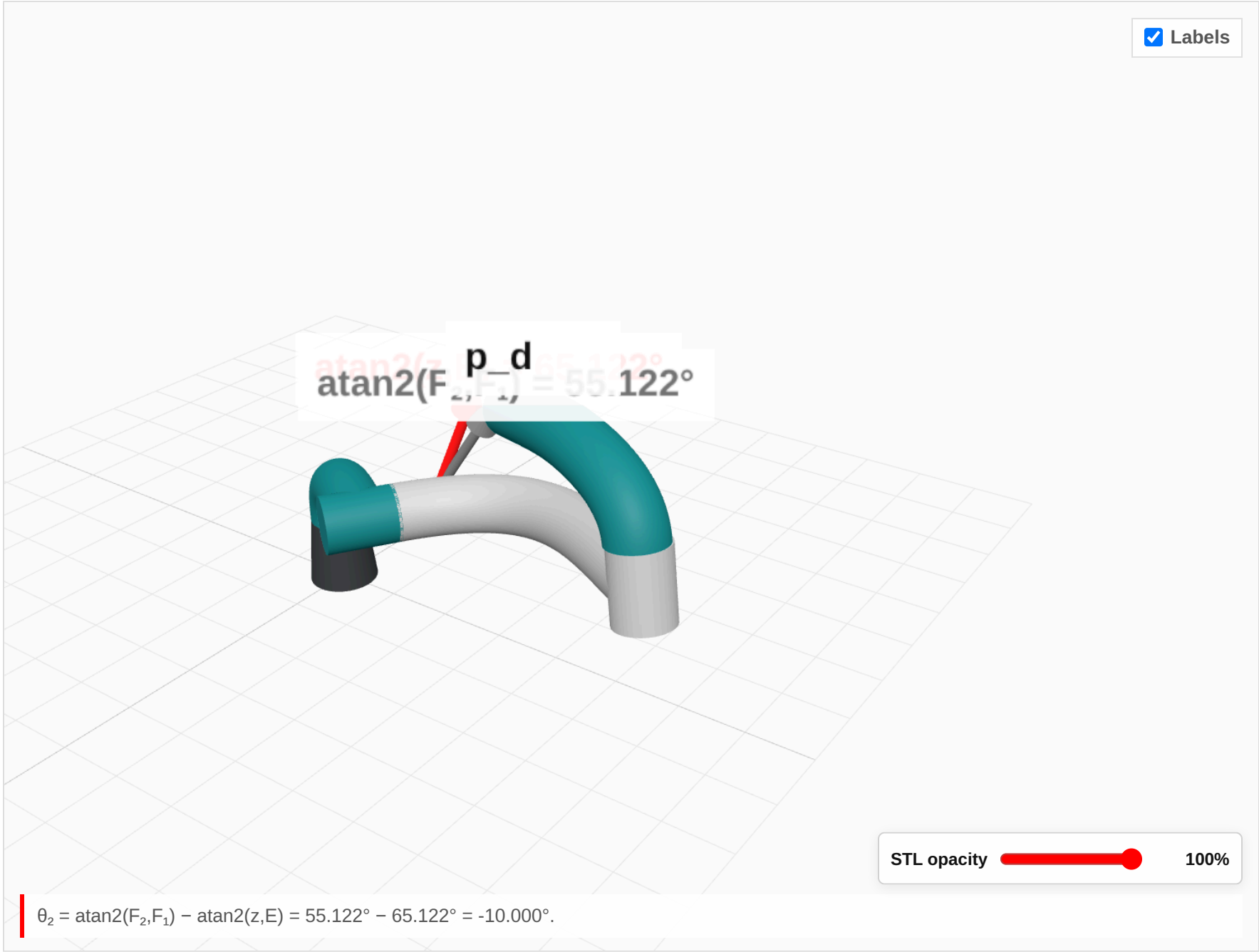
$$\theta_2 = \text{atan } 2(F_2, F_1) - \text{atan } 2(z, E)$$

This is the same angle obtained from $\text{atan } 2(s_2, c_2)$ after Cramer’s rule.

The minus sign follows from $\alpha_1 = -90^\circ$: positive rotation about joint 2 is clockwise in this x – z coordinate plot.

Numerical recovery of θ_2

Select $\theta_3 = -170^\circ$. Evaluate F_1, F_2, F_3 , then solve the same invariant equations by Cramer's rule.



$$\begin{aligned} F_1 &= 0.522788, & F_2 &= 0.75, \\ F_3 &= 2.814973, & E &= 0.384610, \\ z &= 0.829387, & \Delta_2 &= 0.835808. \end{aligned}$$

$$\begin{aligned} c_2 &= \frac{F_1 E + F_2 z}{\Delta_2} = 0.984808, \\ s_2 &= \frac{F_2 E - F_1 z}{\Delta_2} = -0.173648, \\ \theta_2 &= \text{atan } 2(s_2, c_2) = -10^\circ. \end{aligned}$$

The scene shows the equivalent planar-angle difference: $55.122^\circ - 65.122^\circ = -10^\circ$. Values are rounded.

Substitute θ_2, θ_3 , then solve the x, y equations

Set $A = a_1 + F_1 c_2 + F_2 s_2 = a_1 + E$ and $C = d_2 + a_3 s_3$. Only c_1, s_1 remain unknown.

$$\begin{bmatrix} A & -C \\ C & A \end{bmatrix} \begin{bmatrix} c_1 \\ s_1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Cramer's rule gives

$$c_1 = \frac{Ax + Cy}{A^2 + C^2}, \quad s_1 = \frac{Ay - Cx}{A^2 + C^2}.$$

$$\theta_1 = \text{atan2}(s_1, c_1)$$

Equivalently, compare horizontal polar angles:

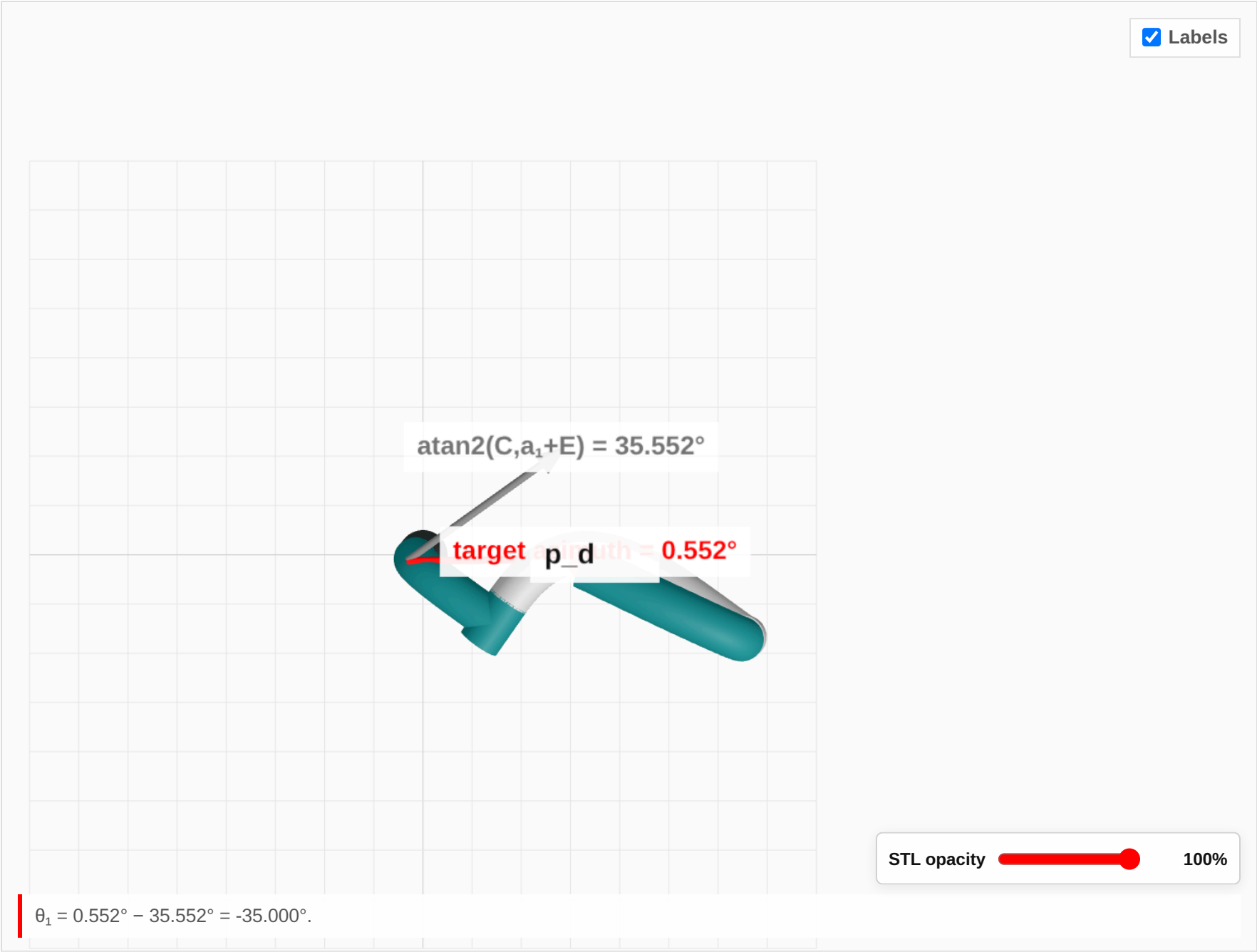
$$\theta_1 = \text{atan2}(y, x) - \text{atan2}(C, A).$$

If $A^2 + C^2 = 0$, the target is on axis 1 and these equations cannot determine θ_1 .

For each real θ_3 , recover θ_2 , then θ_1 , and validate the full world position $[x, y, z_d]^T$ and joint limits.

Numerical recovery of θ_1

The first-joint angle is the azimuth correction between the pre-rotation vector and the target projection.



$$\begin{aligned} A &= 1 + E = 1.384610, \\ C &= 1.25 + 1.5s_3 = 0.989528, \\ \Delta_1 &= A^2 + C^2 = 2.896310. \end{aligned}$$

$$\begin{aligned} c_1 &= \frac{Ax + Cy}{\Delta_1} = 0.819152, \\ s_1 &= \frac{Ay - Cx}{\Delta_1} = -0.573576, \\ \theta_1 &= \text{atan2}(s_1, c_1) = -35^\circ. \end{aligned}$$

In top view, this is the azimuth correction
 $0.552^\circ - 35.552^\circ = -35^\circ$. Values are rounded.

Feasibility tests and the intersecting-axis limit

The coefficient formulation identifies singular recovery cases explicitly; intersecting first and second axes change how we calculate E .

$$a_1 = 0$$

The common-normal distance between axes 1 and 2 vanishes. For the nonparallel axes in this model, that means the axes intersect; coincident axes require a separate case.

$$E = \frac{R - F_3}{2a_1} \text{ is undefined.}$$

Interpretation: the distance equation becomes $R = F_3$, independent of θ_2 . Recover $E = \pm \sqrt{F_1^2 + F_2^2 - z^2}$, then reuse Cramer's rule.

$$F_1^2 + F_2^2 = 0$$

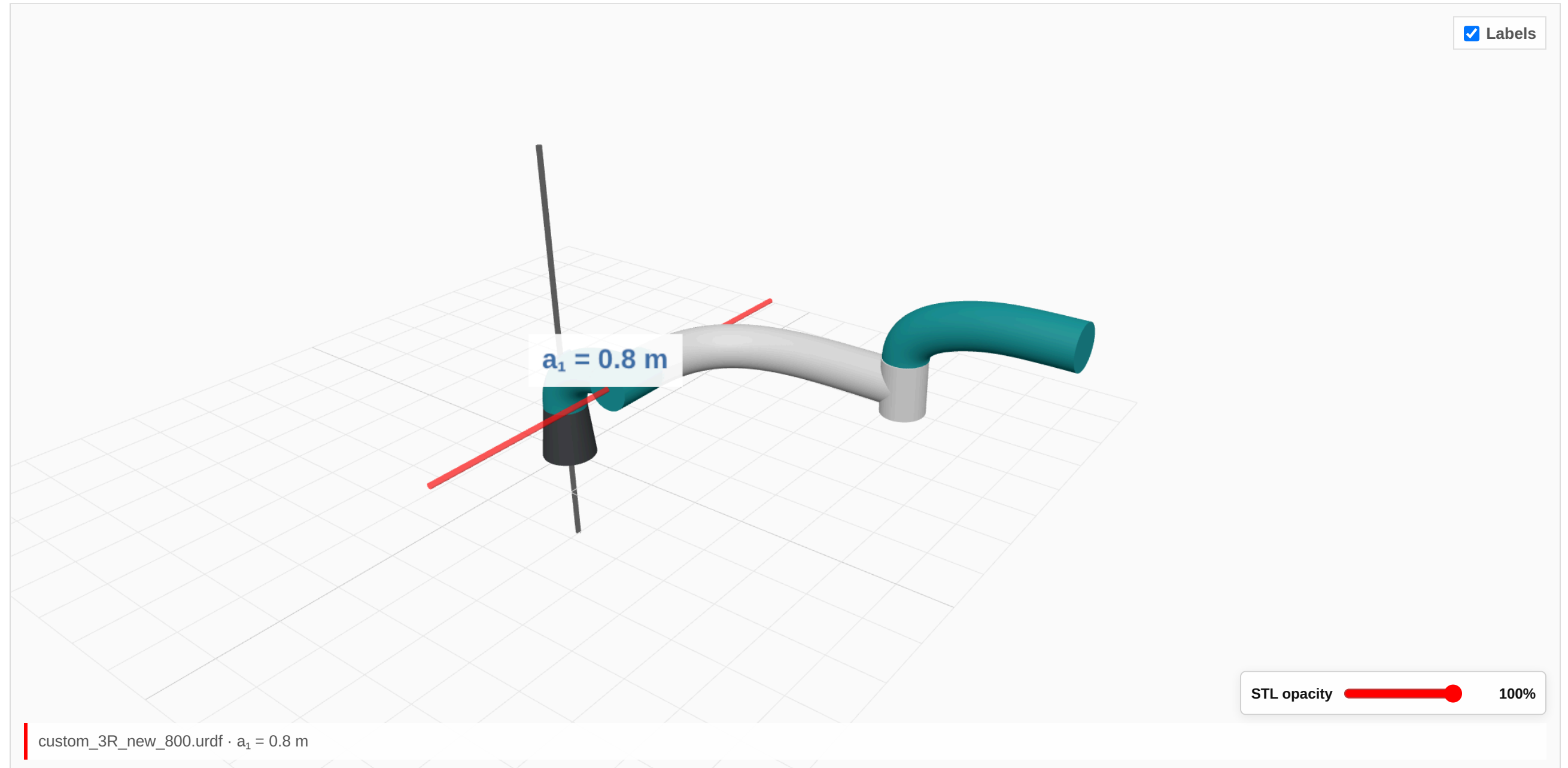
The coefficient matrix for recovering c_2, s_2 loses rank.

$$\det \begin{bmatrix} F_1 & F_2 \\ F_2 & -F_1 \end{bmatrix} = -(F_1^2 + F_2^2).$$

Conclusion: θ_2 is not identifiable from these two invariant equations.

Intersecting axes in the limit $a_1 \rightarrow 0$

A family of related URDF models contracts the common-normal distance a_1 from the offset-axis robot to the intersecting-axis robot.



The ordered-elimination algorithm

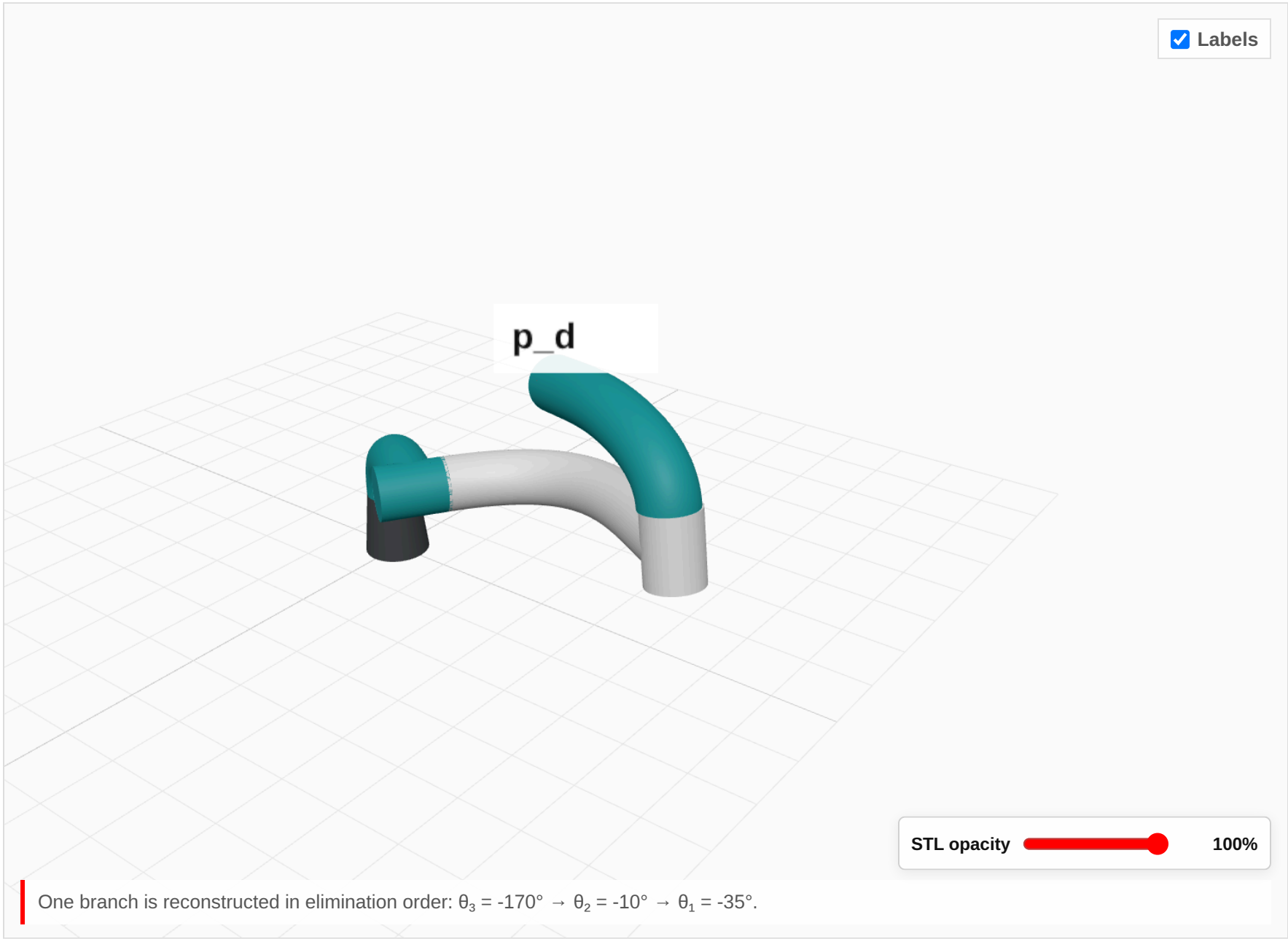
The analytical procedure reduces position IK to one quartic solve, two angle recoveries, and forward-kinematics validation.

- 1 Compute $z = z_d - d_1$, $\rho^2 = x^2 + y^2$, and $R = \rho^2 + z^2$.
- 2 Express R and z using $F_i(\theta_3)$, then set $E = (R - F_3)/(2a_1)$.
- 3 Enforce $E^2 + z^2 = F_1^2 + F_2^2$; substitute $t_3 = \tan(\theta_3/2)$ and solve the quartic. Check $\theta_3 = \pi$ separately.
- 4 Use Cramer's rule on the E, z system for c_2, s_2 , then on the x, y system for c_1, s_1 . Recover both angles with atan2 .
- 5 Apply joint limits and validate every candidate by forward kinematics.

Branch condition: these equations use the custom robot's $\alpha_1 = -90^\circ$, $\alpha_2 = 90^\circ$, with $a_1 \neq 0$. Handle zero recovery determinants separately; intersecting first axes admit a different geometric reduction.

Numerical reconstruction of one IK branch

Back-substitution recovers θ_2 and θ_1 from a selected real root θ_3 , reproducing the target to numerical precision.



target	$\mathbf{p}_d = (1.7018, 0.0164, 1.8294) \text{ m}$
quartic	$\theta_3 = -170.000^\circ$
backprop	$\theta_2 = -10.000^\circ$
azimuth	$\theta_1 = -35.000^\circ$
residual	$\ \mathbf{p}(\mathbf{q}) - \mathbf{p}_d\ < 10^{-12} \text{ m}$

Three other valid branches:

- $(-14.656^\circ, -6.606^\circ, -147.608^\circ),$
- $(178.402^\circ, -147.533^\circ, -61.150^\circ),$
- $(-69.098^\circ, -61.976^\circ, 166.678^\circ).$

SECOND CASE STUDY · INTERSECTING-AXIS GEOMETRY

Paden–Kahan Subproblems

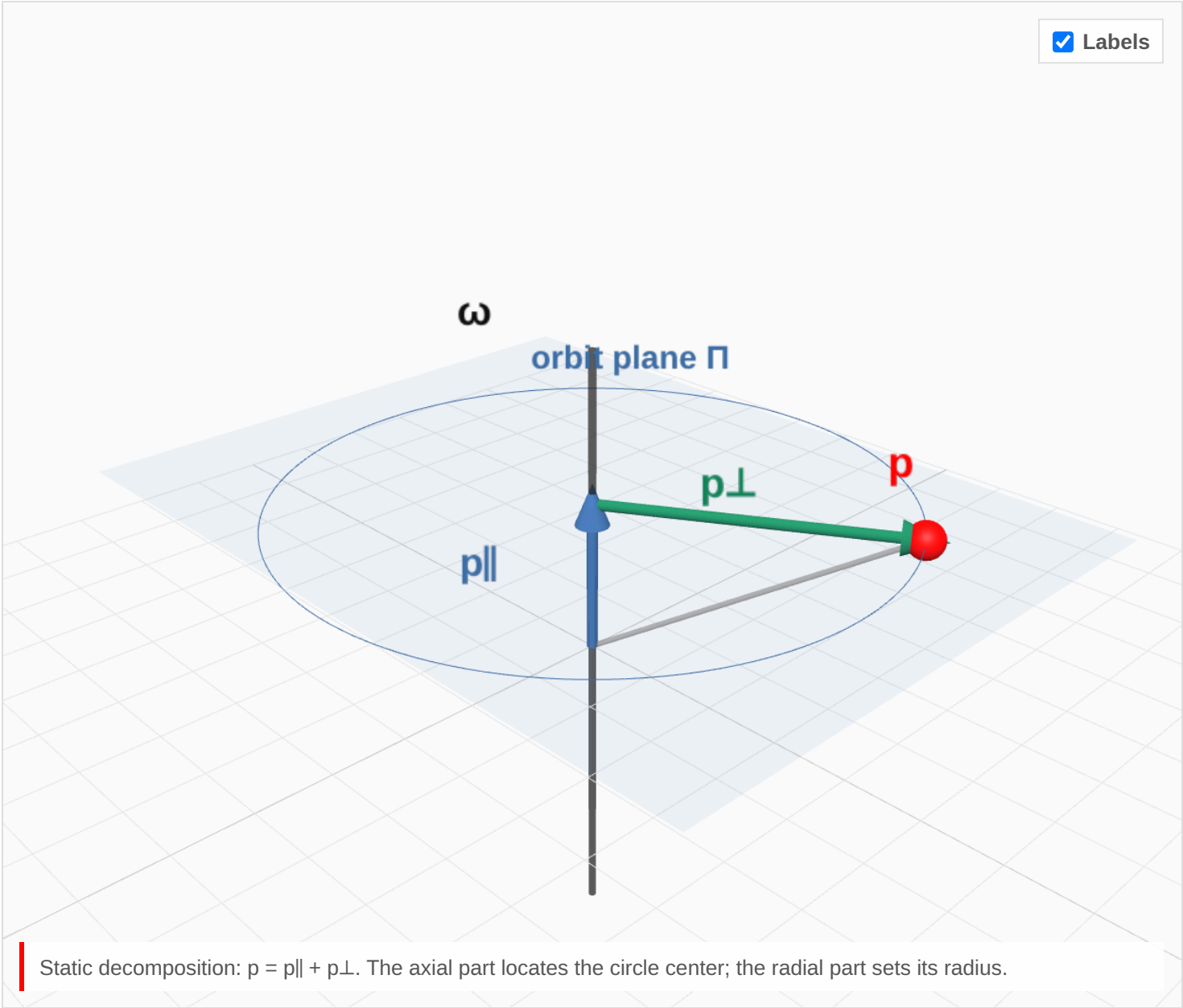
A revolute joint draws a circle. IK asks where that circle meets another constraint.

Circular orbits of revolute motion

Keep $\mathbf{p}$ fixed and decompose it into components along and perpendicular to the unit axis $\boldsymbol{\omega}$. The sheet shows its orbit plane.

$$\begin{aligned}\mathbf{p}_{\parallel} &= (\boldsymbol{\omega}^T \mathbf{p}) \boldsymbol{\omega}, \\ \mathbf{p}_{\perp} &= \mathbf{p} - \mathbf{p}_{\parallel}, \\ \mathbf{p} &= \mathbf{p}_{\parallel} + \mathbf{p}_{\perp}.\end{aligned}$$

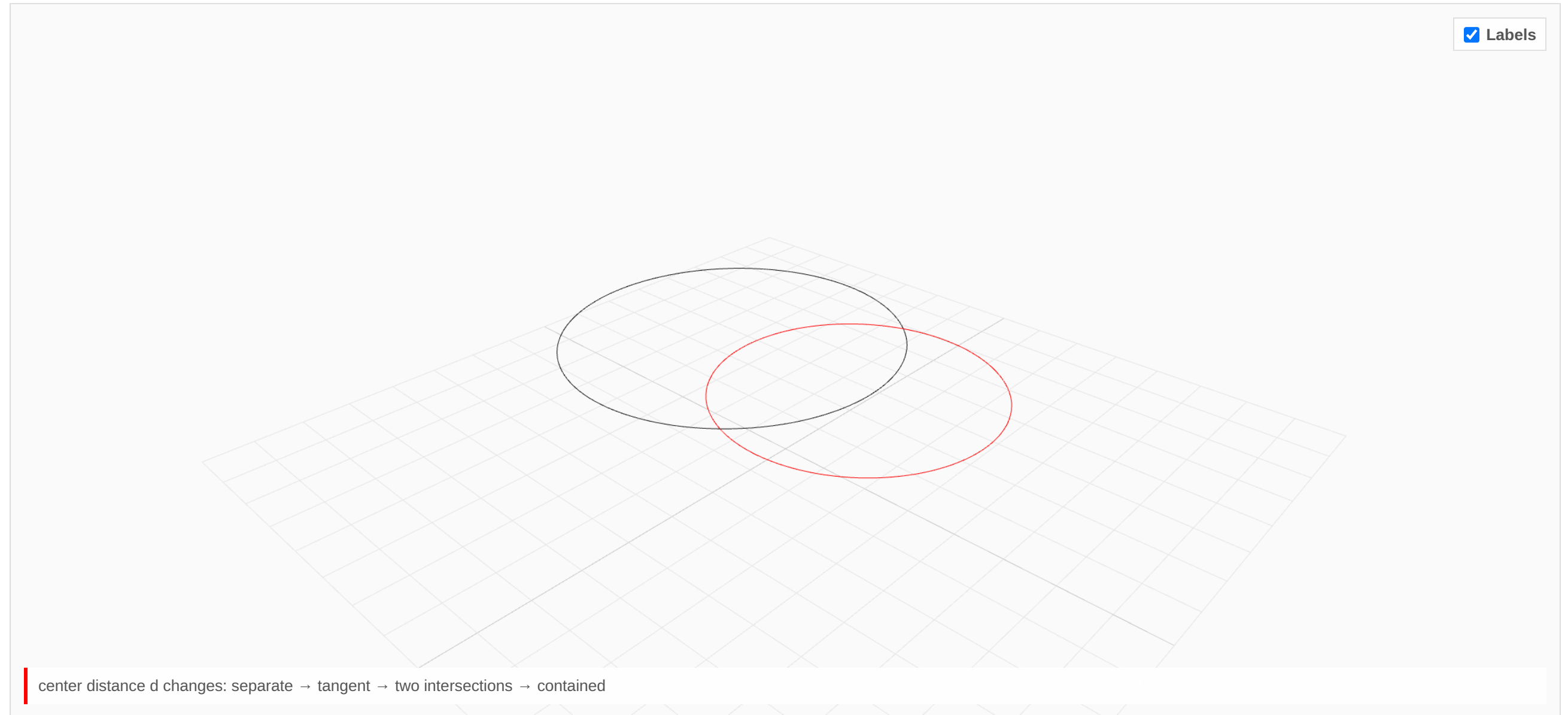
center	$\mathbf{p}_{\parallel}$
radius	$r = \ \mathbf{p}_{\perp}\ $
plane	$\boldsymbol{\omega}^T \mathbf{x} = \boldsymbol{\omega}^T \mathbf{p}$



Rotation preserves axial coordinate, distance to the axis, and Euclidean length.

Circle intersection and geometric feasibility

For coplanar circles with radii r_1, r_2 and center distance d , feasibility is determined by three lengths.



$d > r_1 + r_2$ **or** $d < |r_1 - r_2|$
no intersection

$d = r_1 + r_2$ **or**
 $d = |r_1 - r_2| > 0$
one tangent point

$|r_1 - r_2| < d < r_1 + r_2$
two intersections

$d = 0, r_1 = r_2 > 0$
coincident circles: infinitely many points

Paden–Kahan subproblem 1: point registration

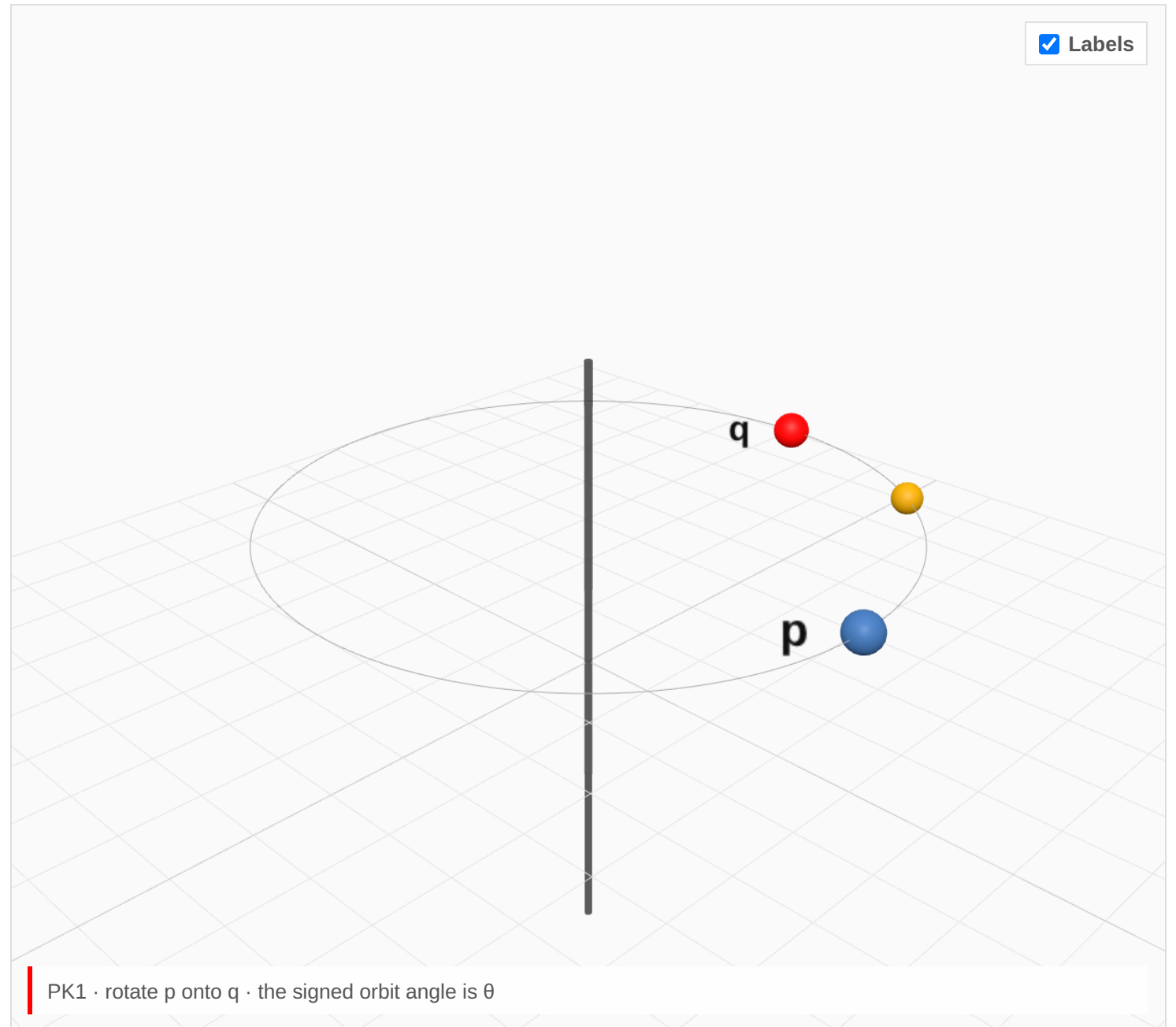
Given unit axis ω and points $\mathbf{p}$, $\mathbf{q}$, find θ such that $R(\omega, \theta)\mathbf{p} = \mathbf{q}$.

PK1

$$R(\omega, \theta)\mathbf{p} = \mathbf{q}$$

Exact feasibility requires

$$\omega^T \mathbf{p} = \omega^T \mathbf{q}, \quad \|\mathbf{p}_\perp\| = \|\mathbf{q}_\perp\|.$$



If feasible and the orbit radius is nonzero, PK1 has one solution modulo 2π .

Cosine measures alignment; sine measures signed rotation

For a feasible PK1 problem, the projected vectors have the same nonzero length r . The unit axis $\boldsymbol{\omega}$ sets the positive rotation direction.

Build coordinates in the orbit plane

$$\begin{aligned}\mathbf{p}_\perp &= (I - \boldsymbol{\omega}\boldsymbol{\omega}^T)\mathbf{p}, \\ \mathbf{q}_\perp &= (I - \boldsymbol{\omega}\boldsymbol{\omega}^T)\mathbf{q}, \\ \mathbf{e}_1 &= \mathbf{p}_\perp / r, \quad \mathbf{e}_2 = \boldsymbol{\omega} \times \mathbf{e}_1.\end{aligned}$$

$\mathbf{e}_1$ points toward the starting point. $\mathbf{e}_2$ is the positive tangent direction given by the right-hand rule.

$$\mathbf{q}_\perp = r \cos \theta \mathbf{e}_1 + r \sin \theta \mathbf{e}_2.$$

Take dot products with these two perpendicular unit vectors to read the coordinates of $\mathbf{q}_\perp$.

Extract cosine and signed sine

$$\begin{aligned}\cos \theta &= \frac{\mathbf{e}_1^T \mathbf{q}_\perp}{r} = \frac{\mathbf{p}_\perp^T \mathbf{q}_\perp}{r^2}, \\ \sin \theta &= \frac{\mathbf{e}_2^T \mathbf{q}_\perp}{r} = \frac{(\boldsymbol{\omega} \times \mathbf{p}_\perp)^T \mathbf{q}_\perp}{r^2} \\ &= \frac{\boldsymbol{\omega}^T (\mathbf{p}_\perp \times \mathbf{q}_\perp)}{r^2}.\end{aligned}$$

The dot product gives alignment. The scalar triple product gives signed area: positive when rotation from $\mathbf{p}_\perp$ to $\mathbf{q}_\perp$ follows $+\boldsymbol{\omega}$.

$$\theta = \text{atan2}(\boldsymbol{\omega}^T (\mathbf{p}_\perp \times \mathbf{q}_\perp), \mathbf{p}_\perp^T \mathbf{q}_\perp).$$

The common positive factor r^2 cancels in atan2 , which keeps the sign and quadrant.

Check: $\boldsymbol{\omega} = \mathbf{e}_z$, $\mathbf{p}_\perp = \mathbf{e}_x$, $\mathbf{q}_\perp = \mathbf{e}_y$ gives $(\cos \theta, \sin \theta) = (0, 1)$, so $\theta = +90^\circ$. If $r = 0$, the point does not move: the angle is arbitrary or the target is infeasible.

Paden–Kahan subproblem 3: a distance constraint

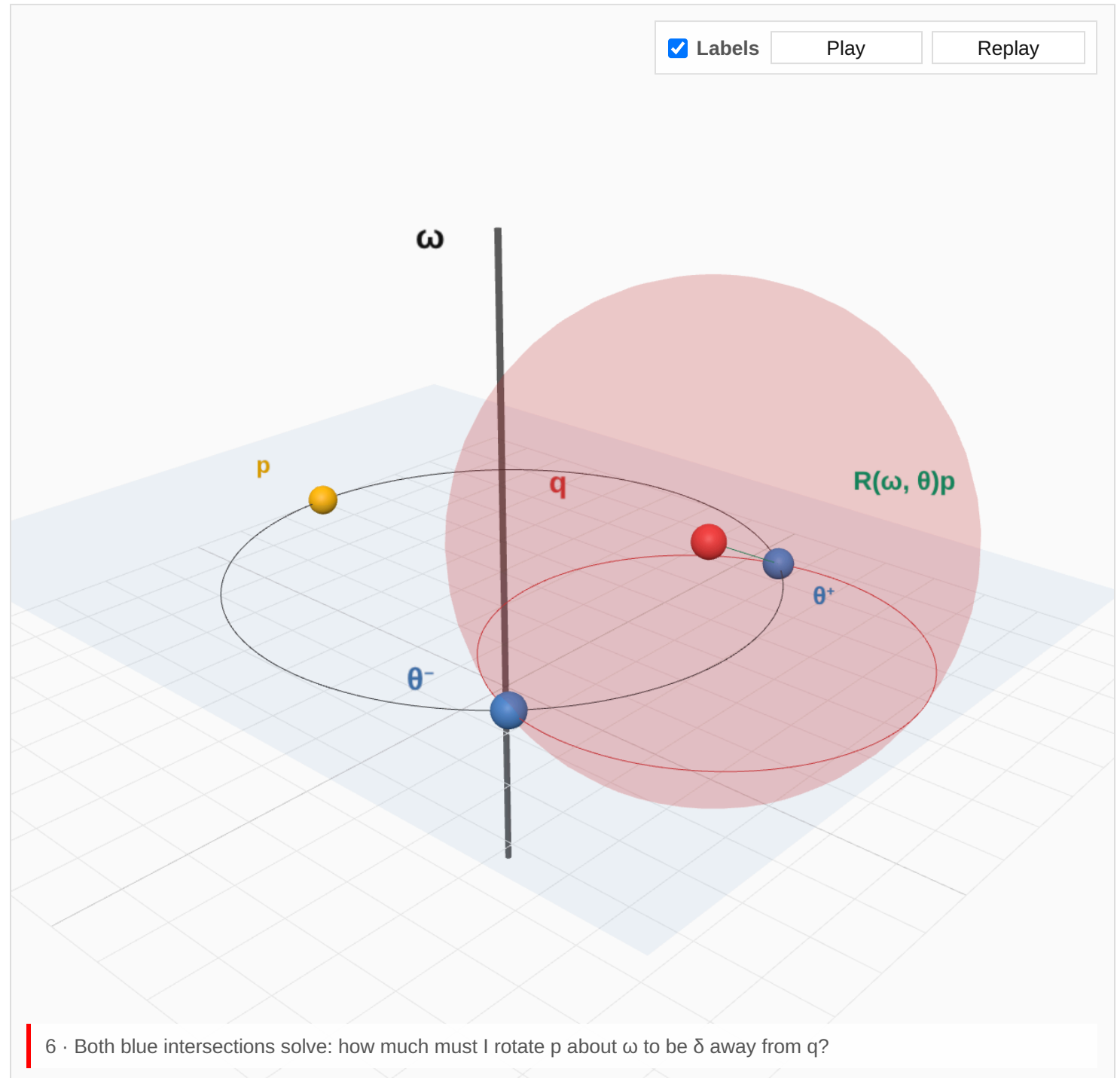
By how much should we rotate $\mathbf{p}$ about $\boldsymbol{\omega}$ so that it ends up exactly δ away from $\mathbf{q}$?

PK3

$$\|R(\boldsymbol{\omega}, \theta)\mathbf{p} - \mathbf{q}\| = \delta$$

The norm is the straight-line distance from the rotated point $R(\boldsymbol{\omega}, \theta)\mathbf{p}$ to $\mathbf{q}$.

- 1 Place $\mathbf{p}$, $\mathbf{q}$, then draw a radius δ from $\mathbf{q}$.
- 2 Sweep that radius to form the sphere of points at distance δ from $\mathbf{q}$.
- 3 Rotate $\mathbf{p}$ to draw its orbit; slice the sphere by this plane and mark the circle intersections.



Planar reduction of the PK3 constraint

Axial separation consumes part of the allowed distance; the remainder becomes an in-plane circle radius.

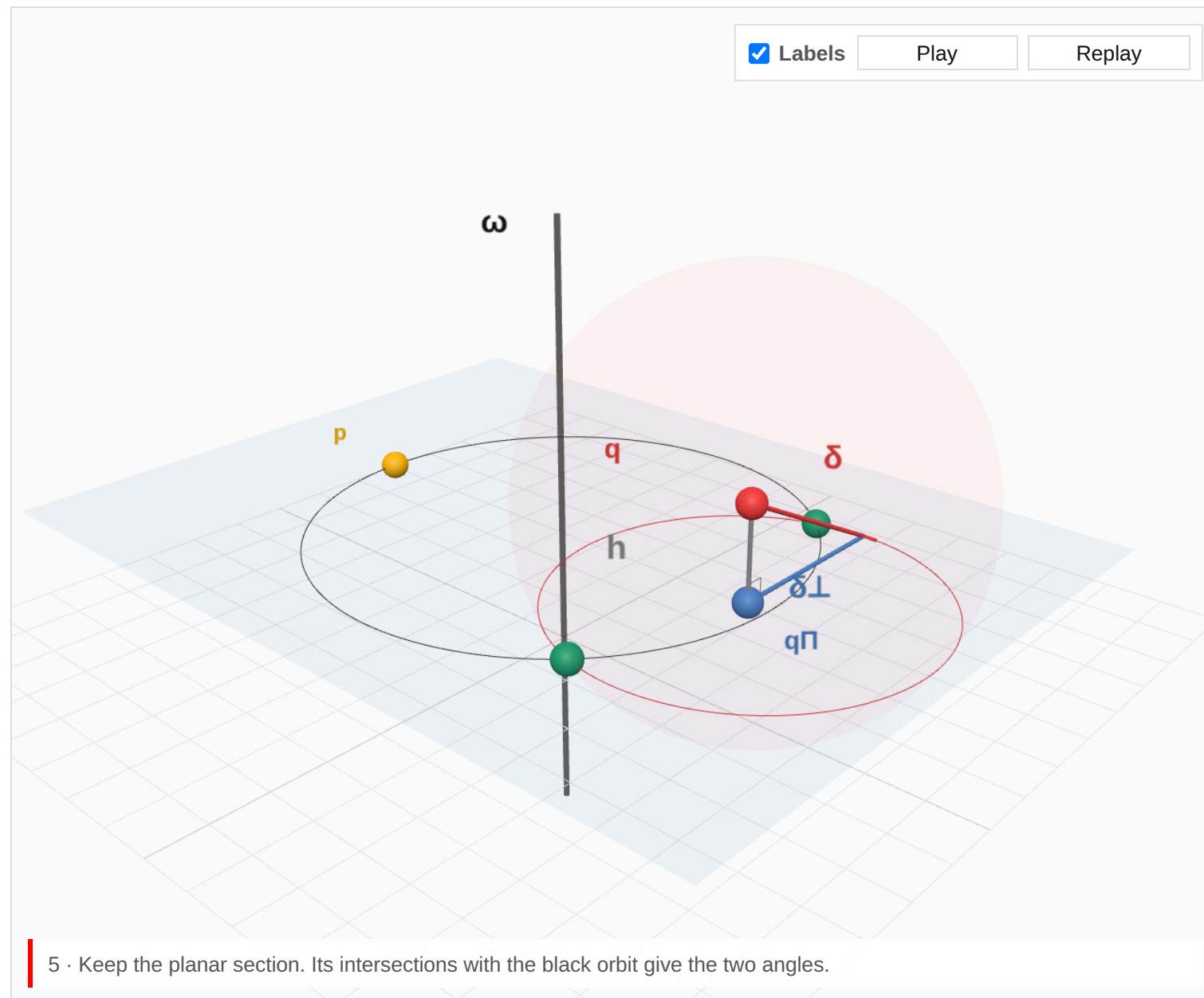
$$\begin{aligned} \mathbf{c} &= (\boldsymbol{\omega}^T \mathbf{p}) \boldsymbol{\omega}, \quad r = \|\mathbf{p} - \mathbf{c}\|, \\ h &= \boldsymbol{\omega}^T (\mathbf{q} - \mathbf{c}), \\ \mathbf{q}_\pi &= \mathbf{q} - h \boldsymbol{\omega}, \\ \mathbf{q}_\perp &= \mathbf{q}_\pi - \mathbf{c}, \quad d = \|\mathbf{q}_\perp\|. \end{aligned}$$

For a point $\mathbf{x}$ in the orbit plane, the vertical and planar distances form a right triangle:

$$\begin{aligned} \|\mathbf{x} - \mathbf{q}\|^2 &= h^2 + \|\mathbf{x} - \mathbf{q}_\pi\|^2, \\ \delta_\perp &= \sqrt{\delta^2 - h^2}. \end{aligned}$$

Check $\delta^2 \geq h^2$, then $|r - \delta_\perp| \leq d \leq r + \delta_\perp$.

Shown: $h = 0.8$, $\delta \simeq 1.7443$, and $\delta_\perp = 1.55$.



Slice the sphere at the orbit plane. The section has center $\mathbf{q}_\pi$ and radius $\delta_\perp$; its intersections with the orbit are exactly the original circle–sphere intersections.

Two PK3 branches from the law of cosines

Let γ rotate $\mathbf{p}_\perp$ toward the second circle's center. Let ϕ be the triangle's remaining angle.

$$\gamma = \text{atan2}(\boldsymbol{\omega}^T(\mathbf{p}_\perp \times \mathbf{q}_\perp), \mathbf{p}_\perp^T \mathbf{q}_\perp)$$

$$\phi = \arccos\left(\frac{r^2 + d^2 - \delta_\perp^2}{2rd}\right)$$

$\theta = \gamma \pm \phi$

two	$0 < \phi < \pi$
one	$\phi = 0$ or π : tangent
none	acos argument outside $[-1, 1]$

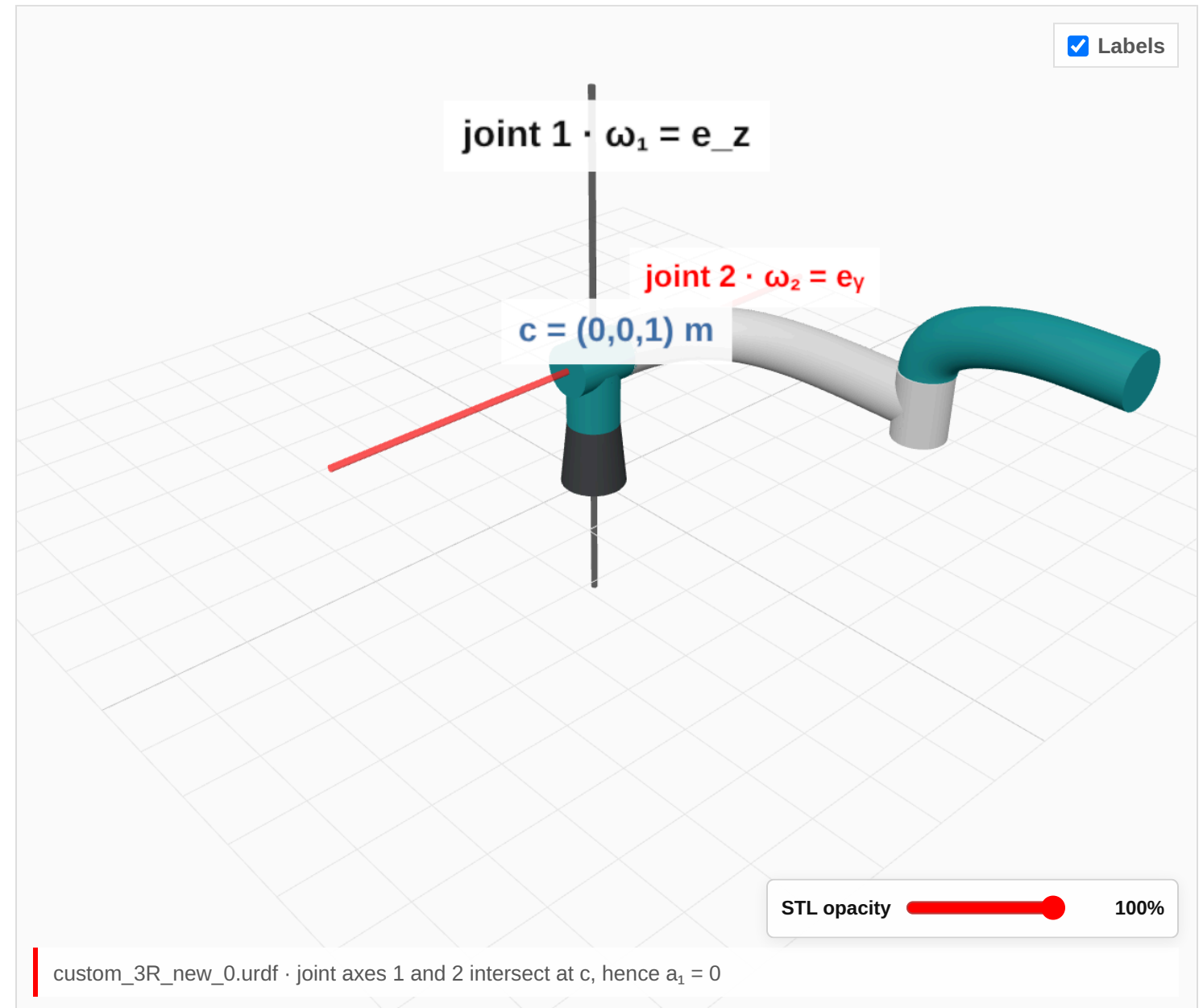
Clamp within floating-point tolerance before `arccos`; reject larger feasibility violations. Handle $r = 0$ or $d = 0$ separately because the formula divides by $2rd$.

A second case study replaces the offset with intersecting axes

custom_3R_new_0 is not the algebraic running example: its joint-1 and joint-2 axes meet at $\mathbf{c} = (0, 0, 1) \text{ m}$, so $a_1 = 0$ and a shorter geometric decomposition becomes available.

$$e^{\hat{\xi}_1 \theta_1} \mathbf{c} = \mathbf{c}, \quad e^{\hat{\xi}_2 \theta_2} \mathbf{c} = \mathbf{c}$$

The black joint-1 axis and red joint-2 axis both pass through the blue point $\mathbf{c}$. Therefore both rotations fix $\mathbf{c}$.



The algebraic degeneracy becomes a geometric advantage.

Robot-specific POE data and position target

This is position IK: only $\mathbf{p}_d$ is prescribed. Tool orientation is deliberately left free, allowing every position branch to survive.

$$\begin{aligned}\boldsymbol{\omega}_1 &= \mathbf{e}_z, & \mathbf{p}_{a,1} &= (0, 0, 0.5), \\ \boldsymbol{\omega}_2 &= \mathbf{e}_y, & \mathbf{p}_{a,2} &= \mathbf{c} = (0, 0, 1), \\ \boldsymbol{\omega}_3 &= \mathbf{e}_z, & \mathbf{p}_{a,3} &= (2, 1.25, 1). \\ \mathbf{p} &= \mathbf{p}_{tool,0} = (3.5, 1.25, 1.75) \text{ m}\end{aligned}$$

shared point $\mathbf{c}$	fixed by joints 1 and 2
tool point $\mathbf{p}$	moved first by joint 3

Numerical target used in the derivation:

$$\mathbf{p}_d = \begin{bmatrix} .882624 \\ .589970 \\ 1.829387 \end{bmatrix} \text{ m}$$

Position equation

$$e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} e^{\hat{\xi}_3 \theta_3} \mathbf{p} = \mathbf{p}_d$$

This target was generated from $(-35^\circ, -10^\circ, -170^\circ)$; the solve below recovers it without using those angles.

Notation and multiplicity: each $e^{\hat{\xi}\theta}$ acts affinely on $[\mathbf{p}; 1]$. PK3 gives up to two θ_3 candidates; for each, up to two real signs of E give angle pairs through the same Cramer formulas.

Build the sphere and the joint-3 orbit

Before moving the robot, convert the target position into the two geometric objects used by PK3.

- 1
- The desired tool position sets a sphere centered at the shared point:

$$\delta = \|\mathbf{p}_d - \mathbf{c}\| = 1.3472 \text{ m}$$
$$\mathcal{S} = \{\mathbf{x} : \|\mathbf{x} - \mathbf{c}\| = \delta\}$$

- 2
- The home tool point is 1.5 m from the third axis:

$$\mathbf{p} = (3.5, 1.25, 1.75), \quad \rho_3 = 1.5 \text{ m}$$
$$\mathcal{C}_3 = \{e^{\hat{\xi}_3 \theta_3} \mathbf{p} : \theta_3 \in [-\pi, \pi]\}$$

PK3 question

Where does $\mathcal{C}_3$ intersect $\mathcal{S}$? $\|e^{\hat{\xi}_3 \theta_3} \mathbf{p} - \mathbf{c}\| = \delta$

The sphere encodes the target distance; the circle encodes every position joint 3 can give the home tool point.

Rotate $\mathbf{p}$ to a circle–sphere intersection

The animation follows the solve order: create the sphere, reveal the robot at its home tool point $\mathbf{p}$, then rotate joint 3 until $\mathbf{p}$ reaches an intersection.

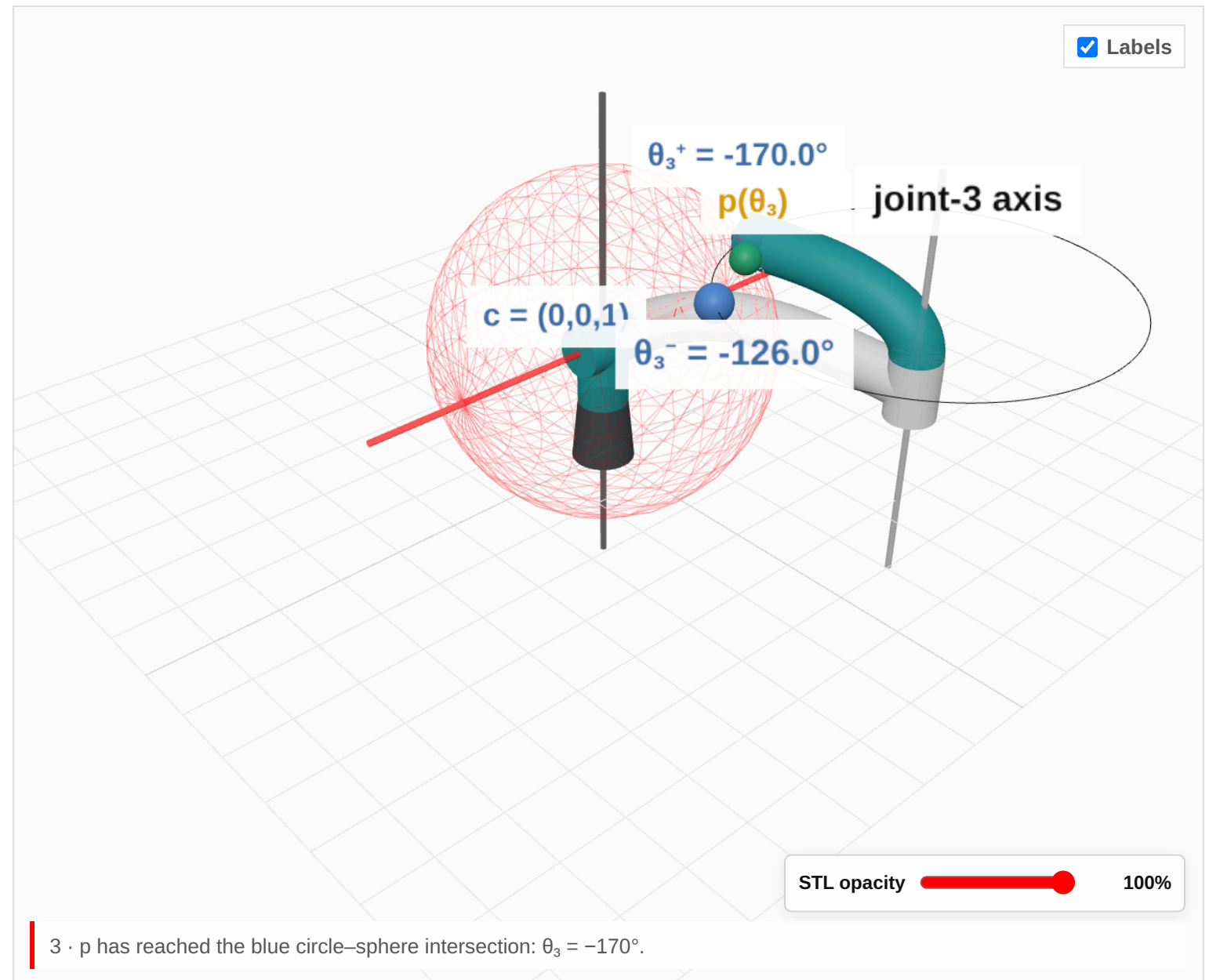
- 1 Sphere:** draw $\|\mathbf{x} - \mathbf{c}\| = \delta$, where $\delta = 1.3472$ m.
- 2 Start:** show the robot at $\theta_3 = 0$, with the tool at $\mathbf{p}$.
- 3 Rotate:** move $\mathbf{p}$ along $\mathcal{C}_3$ until it lies on the sphere.

PK3 scalar equation

$$e^{\hat{\xi}_3 \theta_3} \mathbf{p} - \mathbf{c} = \begin{bmatrix} 2 + 1.5c_3 \\ 1.25 + 1.5s_3 \\ .75 \end{bmatrix}$$

$$\delta^2 = 8.375 + 6c_3 + 3.75s_3$$

$$\theta_3 = -170.000^\circ \text{ or } -125.989^\circ$$



At an intersection, the first two rotations can reorient the vector about $\mathbf{c}$ without changing its length. This is why PK3 determines θ_3 first.

The same Cramer solve, with two possible E values

Once θ_3 is known, reuse slides 26 and 30. Only the calculation of E changes when $a_1 = 0$.

What changes in the distance equation?

$$E = F_1 c_2 + F_2 s_2,$$

$$z = F_2 c_2 - F_1 s_2,$$

$$R = 2a_1 E + F_3.$$

Earlier, $a_1 \neq 0$: the distance equation fixes the sign as well as the magnitude of E .

$$E = \frac{R - F_3}{2a_1}.$$

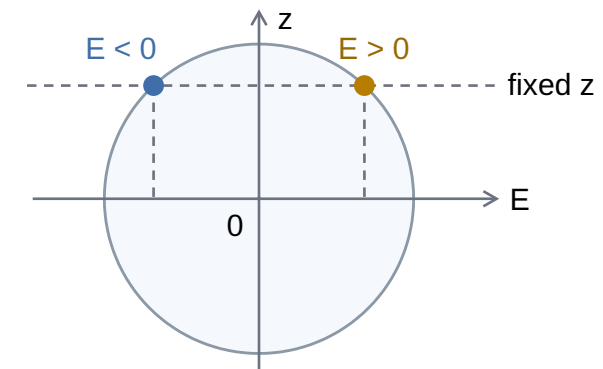
Here, $a_1 = 0$: it becomes $R = F_3(\theta_3)$, independent of θ_2 . It no longer determines E .

Use the unit-circle identity instead

$$\begin{aligned} E^2 + z^2 &= (F_1^2 + F_2^2)(c_2^2 + s_2^2) \\ &= F_1^2 + F_2^2. \end{aligned}$$

$$E = \pm \sqrt{F_1^2 + F_2^2 - z^2}$$

The squared equation determines $|E|$. At the same height z , either sign can satisfy it.



Positive radicand: keep both signs. Zero: the signs merge. Negative: reject the candidate. For each real E , apply the same Cramer formulas as before.

Take $\theta_3 = -170^\circ$, then calculate θ_2

For $\mathbf{p}_d = (.882624, .589970, 1.829387)$ m, use $x = .882624$, $y = .589970$, and shifted height $z = z_d - 1 = .829387$.

1 · Evaluate the known coefficients

$$\begin{aligned} F_1 &= 2 + 1.5 \cos(-170^\circ) = .522788, \\ F_2 &= .750000, \\ C &= 1.25 + 1.5 \sin(-170^\circ) = .989528, \\ \Delta_2 &= F_1^2 + F_2^2 = .835808. \end{aligned}$$

2 · Keep both values of E

$$\begin{aligned} E &= \pm \sqrt{\Delta_2 - z^2} \\ &\simeq \pm \sqrt{.835808 - (.829387)^2} \\ &\simeq \pm .384610. \end{aligned}$$

All displayed values are rounded; calculate using the unrounded coefficients.

3 · Reuse Cramer’s rule from slide 26

$$\begin{aligned} \begin{bmatrix} F_1 & F_2 \\ F_2 & -F_1 \end{bmatrix} \begin{bmatrix} c_2 \\ s_2 \end{bmatrix} &= \begin{bmatrix} E \\ z \end{bmatrix}. \\ c_2 &= \frac{F_1 E + F_2 z}{\Delta_2}, \quad s_2 = \frac{F_2 E - F_1 z}{\Delta_2}, \\ \theta_2 &= \text{atan2}(s_2, c_2). \end{aligned}$$

E	c_2	s_2	θ_2
+.384610	.984808	-.173648	-10.000°
-.384610	.503669	-.863897	-59.757°

Each sign gives one (c_2, s_2) pair and therefore one second-joint angle modulo 2π .

One known θ_3 , two real E values, and the same linear solve twice. Here $F_2 = .75$, so $\Delta_2 > 0$.

Reuse slide 30, with $A = a_1 + E = E$

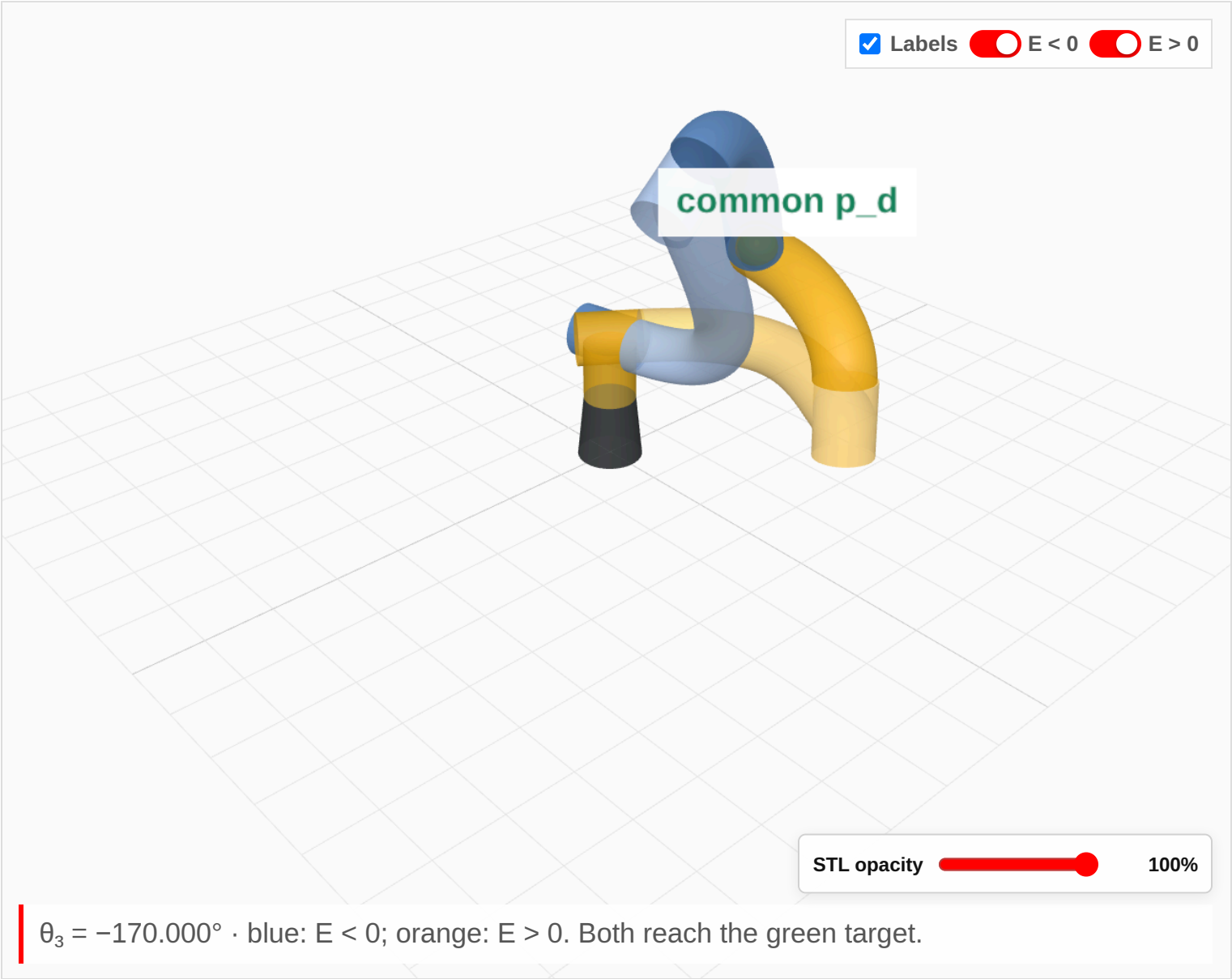
Continue with $\theta_3 = -170^\circ$. Substitute each sign of E into the same first-joint formulas.

$$C = .989528, \quad E = \pm .384610,$$
$$\Delta_1 = E^2 + C^2 = x^2 + y^2 = 1.127090.$$

$$c_1 = \frac{Ex + Cy}{\Delta_1},$$
$$s_1 = \frac{Ey - Cx}{\Delta_1},$$
$$\theta_1 = \text{atan2}(s_1, c_1).$$

E	c_1	s_1	θ_1
> 0	.819152	-.573576	-35.000°
< 0	.216776	-.976221	-77.480°

Thus $(\theta_1, \theta_2, \theta_3)$ is $(-35, -10, -170)^\circ$ or $(-77.480, -59.757, -170)^\circ$.



Both configurations reach the same target. If $\Delta_1 = 0$, the target is on axis 1 and a feasible position leaves θ_1 free.

Repeat for $\theta_3 = -125.989^\circ$

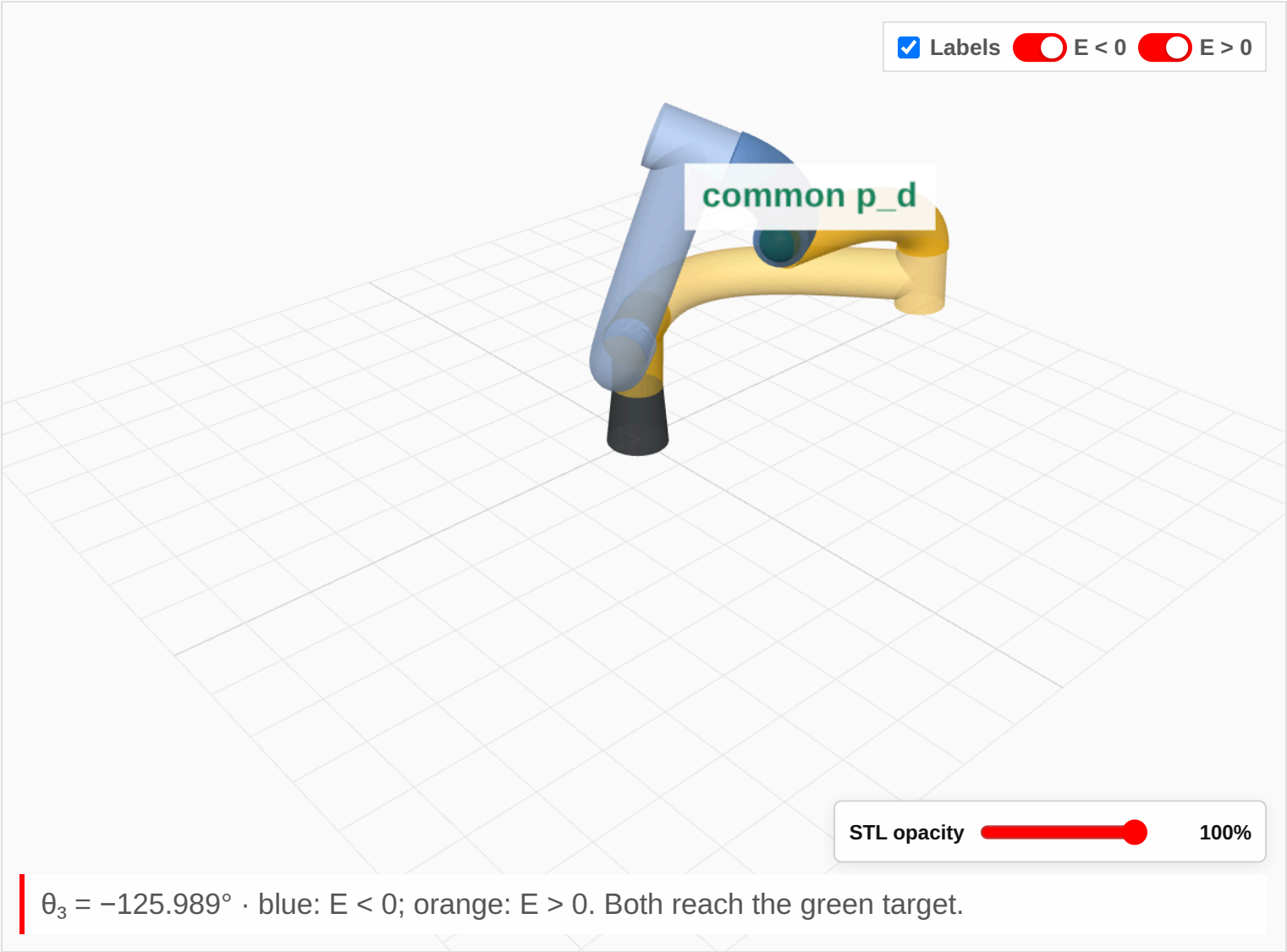
The other PK3 root changes the coefficients. Calculate both E values, then reuse the same Cramer formulas.

$F_1 = 1.118550,$ $F_2 = .750000,$ $C = .036309,$ $\Delta_2 = 1.813654.$

$$E = \pm \sqrt{\Delta_2 - z^2}$$
$$\simeq \pm 1.061024.$$

E	θ_2	θ_1
+1.061024	-4.172°	31.800°
-1.061024	-108.143°	-144.280°

There are two more solutions for the same target. No additional back-substitution formulas are needed.



For each θ_3 : calculate $E = \pm \sqrt{F_1^2 + F_2^2 - z^2}$, recover θ_2 , then recover θ_1 .

Two third-joint roots give four configurations

For this target, both roots give a positive radicand. Each sign of E produces one angle pair through the same Cramer solves.

PK3 two θ_3 values	→	Back-substitute two signs of E per root	→	Total $2 \times 2 = 4$ configurations
E		θ_1	θ_2	θ_3
+ .384610		−35.000°	−10.000°	−170.000°
− .384610		−77.480°	−59.757°	−170.000°
+1.061024		31.800°	−4.172°	−125.989°
−1.061024		−144.280°	−108.143°	−125.989°

Using unrounded angles, forward kinematics of every row gives $\mathbf{p}_d = (.882624, .589970, 1.829387)$ m. Check joint limits before using a configuration.

Algebraic and geometric IK: a comparison

Both approaches eliminate rotations through invariants, but they expose solution structure in different mathematical representations.

D–H ordered elimination

- generic $a_1 \neq 0$, $s_{\alpha 1} \neq 0$;
- invariants eliminate θ_1, θ_2 ;
- a quartic enumerates θ_3 ;
- Cramer's rule recovers θ_2, θ_1 .

Paden–Kahan decomposition

- exploits $a_1 = 0$, intersecting axes;
- fixed points cancel joint motions;
- circle intersections expose feasibility;
- PK3 $\rightarrow$ both real signs of $E \rightarrow$ the same Cramer formulas.

Common principle: eliminate a rotation with something it cannot change—distance, axial projection, or a point on its axis.

TAKEAWAYS

Structural principles of analytical 3R inverse kinematics

Radial invariance, norm preservation, angle back-substitution, and fixed-point geometry provide the main analytical tools.

- 1** Shift $z = z_d - d_1$, then use $R = x^2 + y^2 + z^2$ to eliminate θ_1 .
- 2** Write R and z as coefficient equations in c_2, s_2 ; squaring and adding eliminates θ_2 .
- 3** Cramer's rule recovers each cosine and sine; atan2 gives the joint angle.
- 4** For this $a_1 = 0$ robot, use PK3, calculate both real signs of E , then reuse the same back-substitution: up to four branches.

Further reading: R. Murray, Z. Li & S. Sastry, *A Mathematical Introduction to Robotic Manipulation*, §3.3.

Inverse kinematics as a circle–torus intersection

The IK model through a conformal geometric algebra lens.

$$e^{\theta_1 S_1} e^{\theta_2 S_2} e^{\theta_3 S_3} P = P_d \quad \Longleftrightarrow \quad X \in C_B \cap C_A(\theta_2)$$

Geometric construction adapted from Nayak & Salunkhe, *Inverse kinematic solution for generic 3R positional robots using Conformal Geometric Algebra*.

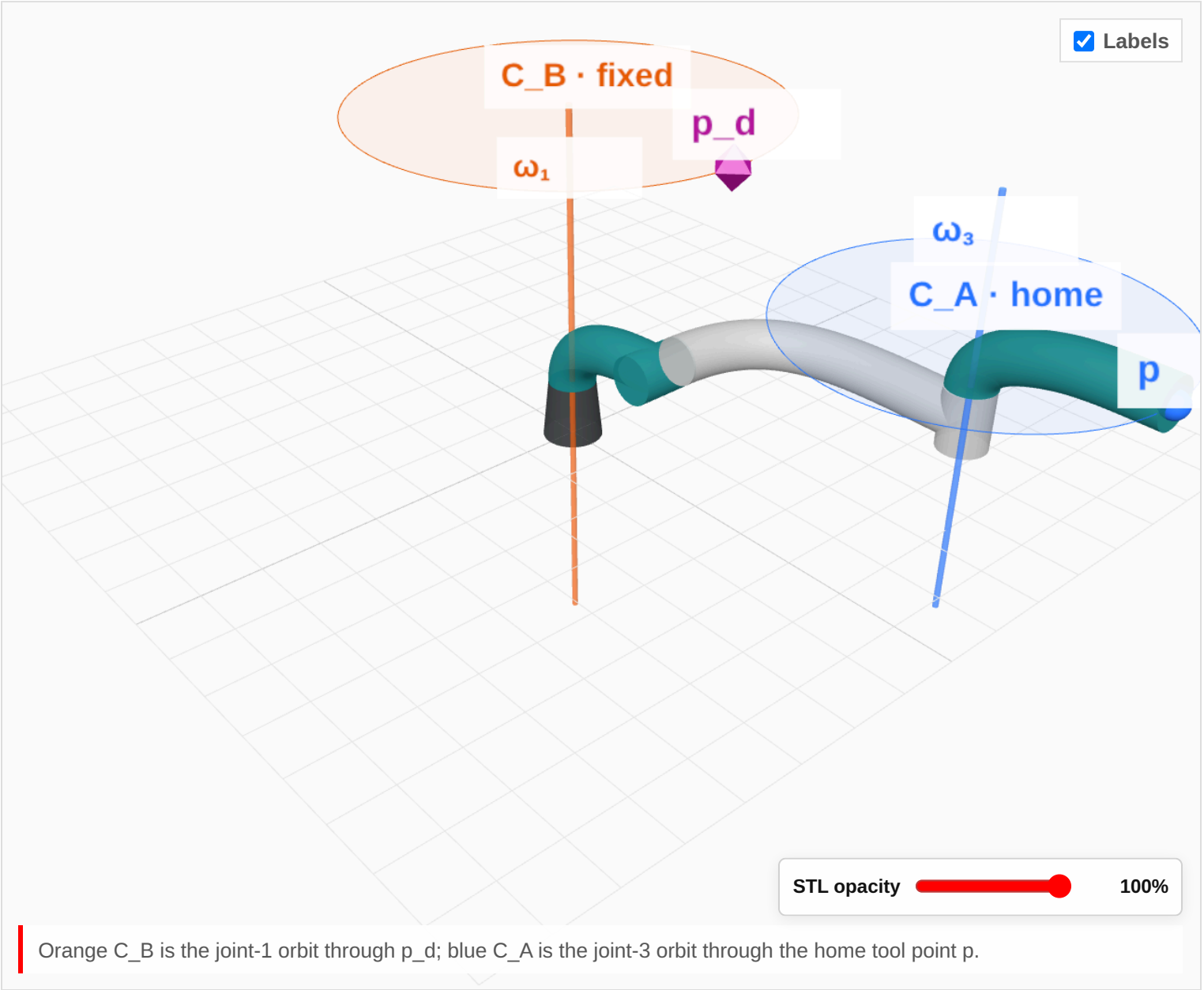
Start with one fixed circle and one moving circle

Both are ordinary Euclidean circles; CGA makes the incidence and rigid-motion operations algebraically uniform.

fixed C_B passes through p_d , is centered on axis 1, and lies in a plane normal to ω_1

home C_A passes through the home tool point p , is centered on axis 3, and lies in a plane normal to ω_3

$$C_B = \text{Orb}_{\omega_1}(p_d), \quad C_A = \text{Orb}_{\omega_3}(p)$$
$$\mathbf{p}_d = (1.7018, 0.0164, 2.8294) \text{ m}$$



The target circle contains every point that joint 1 can rotate into p_d ; the home circle contains every point joint 3 can reach from p .

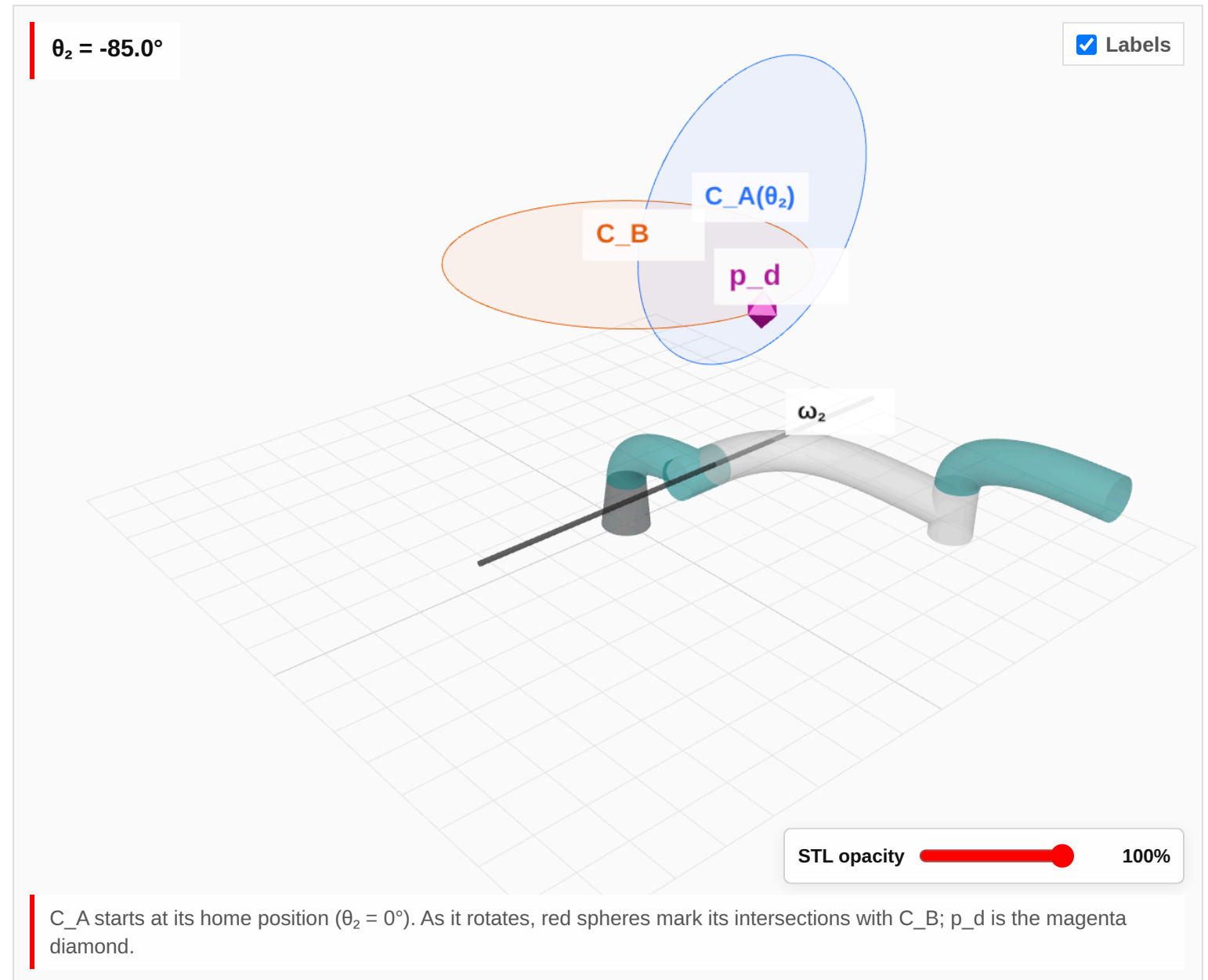
Rotate the home circle about axis 2

Start at $\theta_2 = 0$, where C_A is at home. Watch the displayed angle change as the circle rotates; red spheres mark intersections, while the magenta diamond marks p_d .

$$C_A(\theta_2) = \mathcal{R}_2(\theta_2) C_A \tilde{\mathcal{R}}_2(\theta_2)$$

$$X(\theta_2) = C_A(\theta_2) \vee C_B$$

The meet $\vee$ returns a point-pair object. Its factorization and sign determine whether the pair contains two real points, one repeated tangent point, or no real Euclidean points.



No torus is drawn yet: this slide isolates the moving circle and marks each real circle–circle intersection as it occurs.

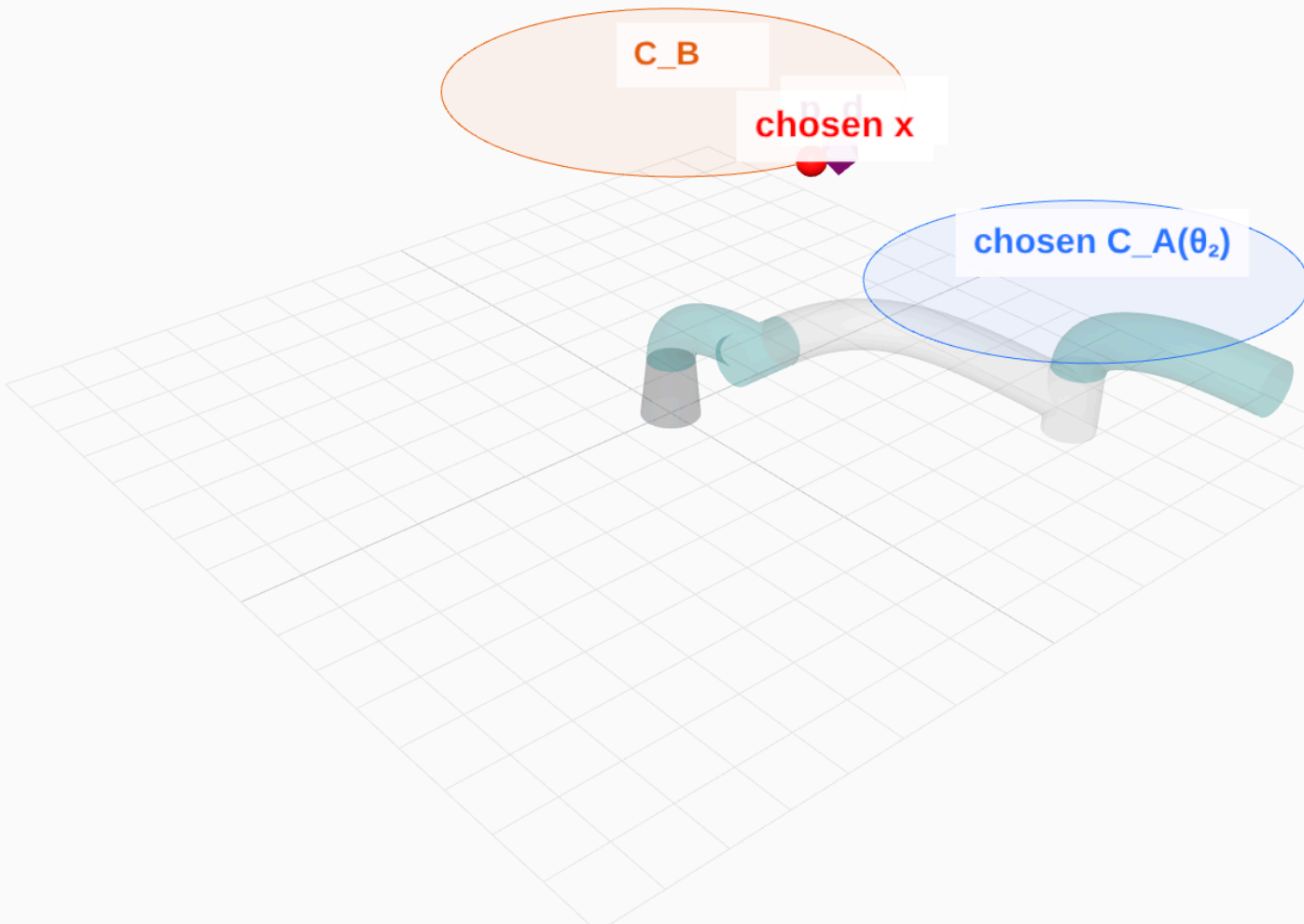
Choose an intersection, then rotate C_A to it

Raising p_d by 1 m leaves two real branches. Use the circle selector in the visualization to replay either one.

branch	θ_2
1	-46.888°
2	-132.273°

$$X_i = R_1(-\theta_{1i})p_d = R_2(\theta_{2i})R_3(\theta_{3i})p$$

☒ Labels Circle IK 1 · θ_2 -46.9° ▼



STL opacity 100%

Choose an IK circle. The blue home circle then rotates by its θ_2 until it meets the selected point on C_B .

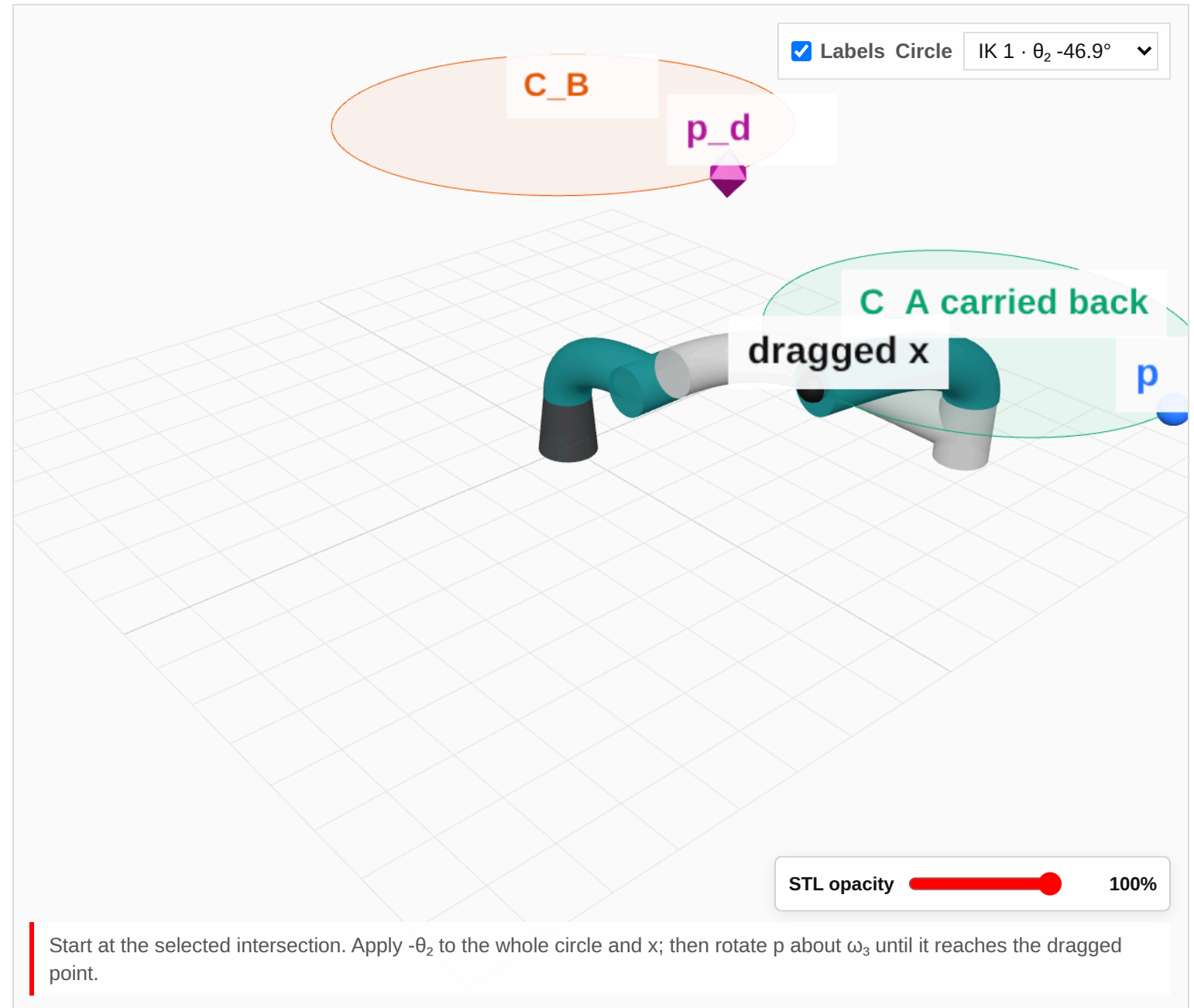
Selecting a branch resets C_A to home and animates it to the corresponding point $x_i \in C_B$.

Undo joint 2, then measure an arc on C_A

Start from either selected IK circle already positioned at its intersection with C_B .

- 1 Take its intersection point $x \in C_B \cap C_A(\theta_2)$.
- 2 Reverse the circle rotation:
 $a = R_2(-\theta_2)x \in C_A$.
- 3 Solve $R_3(\theta_3)p = a$: branch 1 gives
 $\theta_3 = -97.515^\circ$; branch 2 gives -39.698° .

$$R_3(\theta_3)p = R_2(-\theta_2)x$$



The circle and its intersection point move together under $-\theta_2$; only after they reach home does the joint-3 rotation begin.

Reach the intersection, then travel along C_B

Forward motion makes the geometric factorization visible in the same order as the product of rotations.

$$p \xrightarrow{\theta_3} R_3 p \xrightarrow{\theta_2} x \xrightarrow{\theta_1} p_d$$

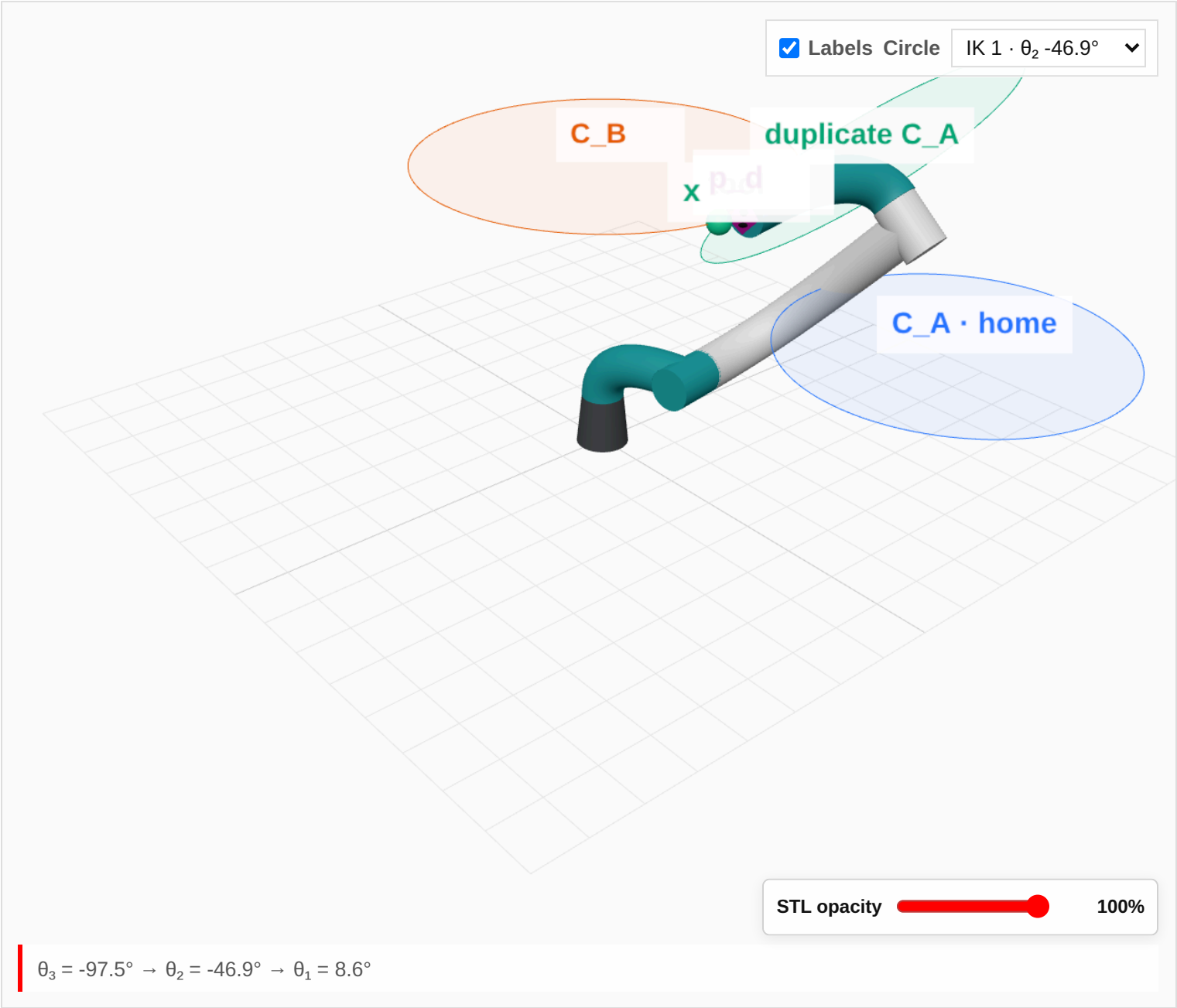
- joint 3

moves p around the home circle C_A
- joint 2

carries that point and circle to the intersection x
- joint 1

keeps the point on C_B and rotates x to p_d

$R_1(8.561^\circ)R_2(-46.888^\circ)R_3(-97.515^\circ)p = p_d$



The final joint-1 motion remains on the fixed orange circle by construction, so it ends exactly at p_d .

Choose p_d , then build an IK branch by dragging

Set the target, then rotate θ_2 , θ_3 , and θ_1 . Follow the instruction inside the scene; the robot moves with the active joint.

- 1 Set p_d . The magenta diamond is the target.
- 2 Rotate θ_2 . It snaps with a click, pauses for 2 seconds, then the circle returns home.
- 3 Rotate θ_3 . It snaps, pauses for 2 seconds, then the circle and robot return to the chosen intersection.
- 4 Rotate θ_1 . Snap to the final angle and reach p_d .

Only the current joint is enabled. Changing the target or choosing “Reset joints” starts a new construction.

θ_2

0.00

θ_3

0.00

θ_1

0.00

Set p_d (magenta diamond), then rotate θ_2 until the circles meet.

☒ Labels

p_x 1.70

p_y 0.02

p_z 2.83

Reset joints

C_B

p_d

$C_A(\theta_2)$

tool

STL opacity 100%

2 IK branches found · snap within 4°

Choose a branch with θ_2 , find θ_3 on the home circle, then use θ_1 to carry the intersection along C_B to the target.

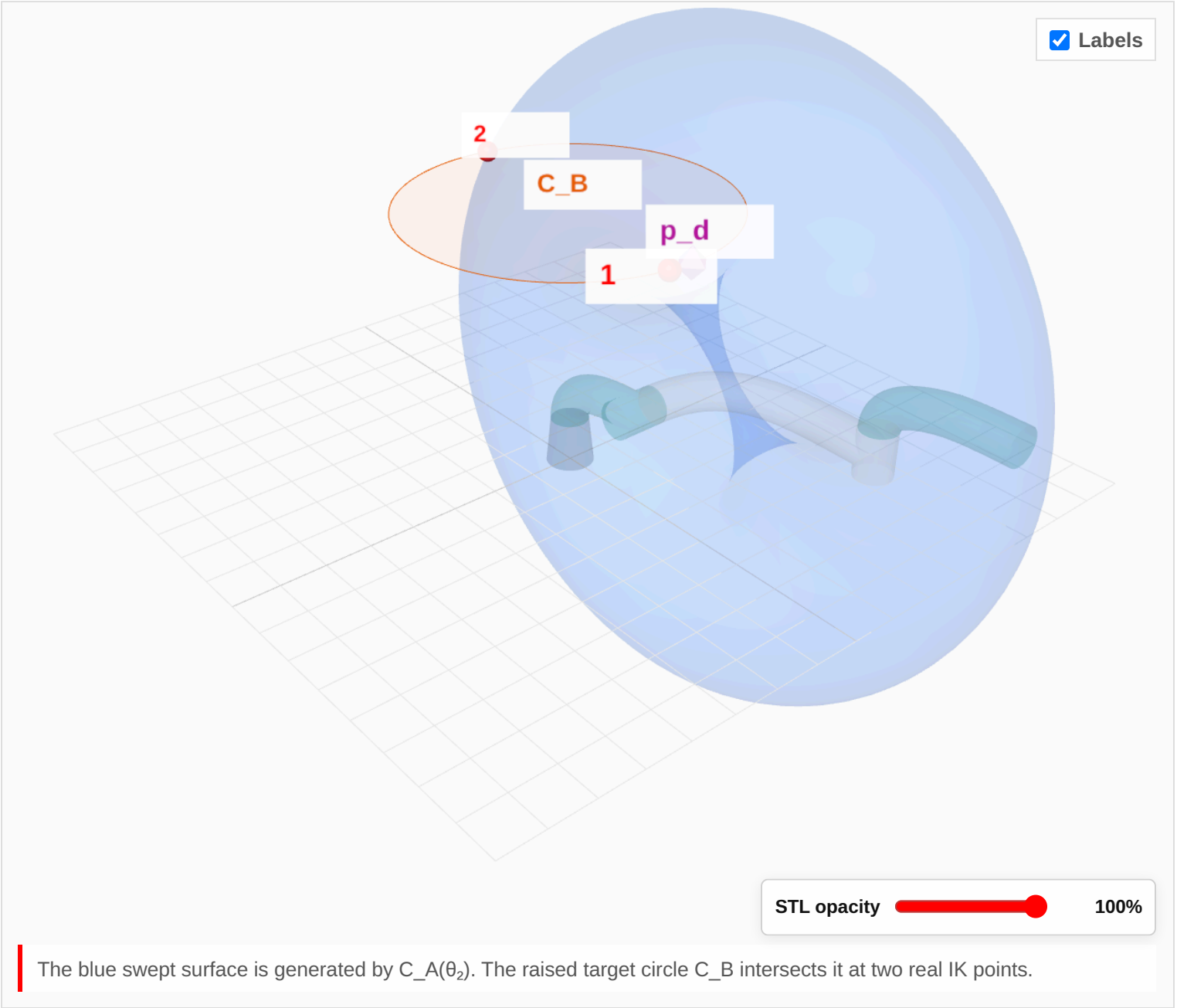
The swept circle is a torus; crossings are IK branches

Let θ_2 span one revolution. For the raised target, C_B cuts the swept surface at two real branch points.

$$\mathcal{T} = \bigcup_{\theta_2 \in [-\pi, \pi)} C_A(\theta_2),$$
$$C_B \cap \mathcal{T} = \{X_1, X_2\}.$$

what CGA adds	points, spheres, planes, circles, meets, and rotors share one algebraic language
scope	the full derivation and quartic expansion are intentionally deferred

For Euclidean 3-space, the paper uses the five-dimensional conformal algebra $\mathcal{G}_{4,1}$, i.e. signature $(4, 1, 0)$.



The point of this preview: the algebra may be advanced, but its geometry is beautifully direct—Ik is an intersection problem.

A 3R position solver becomes the arm half of a 6R pose solver

When the last three axes meet at a spherical wrist, inverse kinematics separates into two coordinated subproblems.

01

Arm

Use 3R position IK to place the wrist center and retain every real arm branch.



02

Wrist

For each arm branch, solve the residual orientation and enumerate its wrist flips.

Next: robot geometry turns one 6R pose problem into a branch tree built from the 3R tools developed here.